§9.1—Intro to Parametric & Vector Calculus

In Algebra, equations are graphed in two variables, x and y. Now we will graph equations with x, y, and t, or with x, y, and θ, where x and y are expressed independently in terms of t or θ. The third variable, t or θ is called the parameter, and the separate equation are called parametric equations.

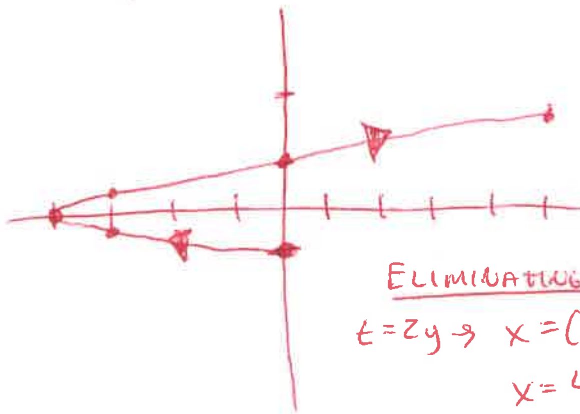
Example 1:
Without a calculator, make a table, and sketch the curve, indicating the direction of your graph. Then eliminate the parameter. Verify on your calculator.

$$x = t^2 - 4 \text{ and } y = \frac{t}{2},\ -2 \le t \le 3$$ SET OF PARAMETRIC EQ.

t	x	y
-2	0	-1
-1	-3	$-\frac{1}{2}$
0	-4	0
1	-3	$\frac{1}{2}$
2	0	1
3	5	$\frac{3}{2}$

$\to (0,-1)$

FASTER HOR THAN VER



t-STEP
SPEED + DIRECTION

ELIMINATING THE PARAMETER

$t = 2y \to x = (2y)^2 - 4$

$x = 4y^2 - 4$

$y \in [-1, 1.5]$

What do you notice about the graphs of $x = 4t^2 - 4$ and $y = t$, $-1 \le t \le 1.5$

What do you notice about the graphs of $x = 4(2\sin t + 1)^2 - 4$ and $y = 2\sin t + 1$, $-1.571 \le t \le 0.253$

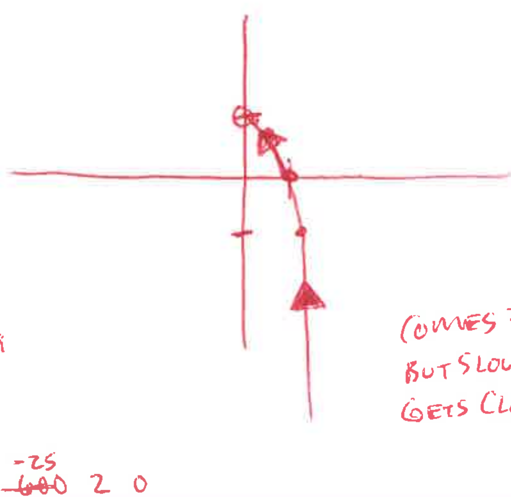
PATH, DIRECTION (SAME)
SPEED A LITTLE FASTER

While rectangular equations on restricted intervals show the PATH, parametric equations show the PATH, DIRECTION, and SPEED.

Example 2:

Do the same for $x = \dfrac{1}{\sqrt{t+1}}$, $y = \dfrac{t}{t+1}$ D: $t > -1$ $\lim\limits_{t \to \infty} \frac{t}{t+1} = 1$

t	x	y
0	1	0
3	$\frac{1}{2}$	$\frac{3}{4}$
8	$\frac{1}{3}$	$\frac{8}{9}$
15	$\frac{1}{4}$	$\frac{15}{16}$
-.1	$\sqrt{2}$	-1



$y(t+1) = t$

$yt + y = t$

$yt - t = -y$

$t(y-1) = -y$

$t = \frac{-y}{y-1}$

$t = \frac{y}{1-y}$

$x = \dfrac{1}{\sqrt{\frac{y}{1-y} + 1}}$

$x^2 = \dfrac{1}{\frac{y}{1-y} + 1} \cdot \dfrac{1-y}{1-y}$

$x^2 = \dfrac{1-y}{y + 1 - y}$

$x^2 = 1 - y$

$\boxed{y = -x^2 + 1}$

$x > 0$

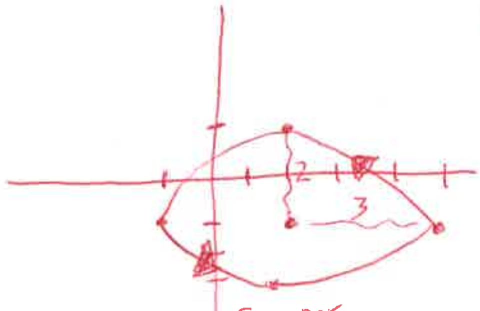
COMES IN VERY FAST
BUT SLOWS DOWN AS IT
GETS CLOSER TO HOLE

CALC
TMIN = -.99
TMAX = 10
TSTEP = .1

-.1 -25
-.25 -600 2 0

Example 3:
Do the same for $x = 2 + 3\cos t$, $y = -1 + 2\sin t$

t	x	y
0	5	-1
$\frac{\pi}{6}$	$2+\frac{3\sqrt{3}}{2}$	0
$\frac{\pi}{2}$	2	1
π	-1	-1
$\frac{3\pi}{2}$	2	-3
2π	5	-1



ELLIPSE

$C(2,-1)$
$x = 3$
$y = 2$

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

$\frac{(x-2)^2}{9} + \frac{(y+1)^2}{4} = 1$

$\cos t = \frac{x-2}{3}$ $\quad \sin t = \frac{y+1}{2}$

$\cos^2 t + \sin^2 t = 1$

$\frac{(x-2)^2}{9} + \frac{(y+1)^2}{4} = 1$

0
2π
.1
-2
6
1
-4
2
1

Time to B.O.T.C.!

Parametric Equations & Formulas for Calculus

If a smooth curve C is given by the equations $x = f(t)$ and $y = g(t)$, then the slope of C at the point (x, y) is given by $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ where $\frac{dx}{dt} \neq 0$, and the second derivative is given by

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d}{dt}\left[\frac{dy}{dx}\right] \cdot \frac{dt}{dx} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}.$$

Example 4:

Without a calculator, given $x = 2\sqrt{t}$, $y = 3t^2 - 2t$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ and evaluate at $t = 1$.

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)} = \frac{6t-2}{t^{-1/2}}$

$= (6t-2)(t^{1/2})$

$= 6t^{3/2} - 2t^{1/2}$

$\left.\frac{dy}{dx}\right|_{t=1} = 4$

$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{9t^{1/2} - t^{-1/2}}{t^{-1/2}}$

$= (9t^{1/2} - t^{-1/2})(t^{1/2})$

$= 9t - 1$

$\left.\frac{d^2y}{dx^2}\right|_{t=1} = 8$

Example 5:
Without a calculator, given $x = 4\cos t$, $y = 3\sin t$, write an equation of the tangent line to the curve at the point where $t = \frac{3\pi}{4}$.

$\left(x\left(\frac{3\pi}{4}\right), y\left(\frac{3\pi}{4}\right)\right)$

$\left(-2\sqrt{2}, \frac{3\sqrt{2}}{2}\right)$

$\frac{dy}{dx} = \frac{3\cos t}{-4\sin t}$

$\left.\frac{dy}{dx}\right|_{t=\frac{3\pi}{4}} = \left(\frac{-3}{4}\right)\frac{(-\sqrt{2}/2)}{(+\sqrt{2}/2)}$

$= \frac{3}{4}$

EQ: $\boxed{L(x) = \frac{3\sqrt{2}}{2} + \frac{3}{4}\left(x + 2\sqrt{2}\right)}$

Example 6:
Without a calculator, find all points of horizontal and vertical tangency given $x = t^2 + t$, $y = t^2 - 3t + 5$.

$\frac{dy}{dx} = \frac{2t-3}{2t+1}$

HORIZ TANG: $\frac{0}{\neq 0} \rightarrow 2t-3=0,\ t=\frac{3}{2}$; So $\left(\left(\frac{3}{2}\right)^2+\frac{3}{2}, \left(\frac{3}{2}\right)^2-3\left(\frac{3}{2}\right)+5\right)$

VERT TANG: $\frac{\neq 0}{0} \rightarrow 2t+1=0,\ t=\frac{-1}{2}$; So $\left(\left(-\frac{1}{2}\right)^2-\frac{1}{2}, \left(-\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)+5\right)$

Parametric Arc Length

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$ is the length of the arc from $t = a$ to $t = b$

$d = \sqrt{\Delta x^2 + \Delta y^2}$

Example 7:
Without a calculator, find the arc length of the given curve if $x = t^2$, $y = 4t^3 - 1$, $0 \le t \le 1$.

$x'(t) = 2t$, $y'(t) = 12t^2$

$$L = \int_0^1 \sqrt{(2t)^2 + (12t^2)^2}\,dt$$

$$= \int_0^1 \sqrt{4t^2 + 144t^4}\,dt$$

$$= \int_0^1 \sqrt{4t^2(1+36t^2)}\,dt$$

$$= \int_0^1 2t\sqrt{1+36t^2}\,dt$$

$$= 2\left(\frac{1}{72}\right)\left(\frac{2}{3}\right)\left(1+36t^2\right)^{3/2}\Big|_0^1$$

$$\boxed{= \frac{1}{54}\left[37^{3/2} - 1^{3/2}\right]}$$

It's time to revisit particle motion.

Horizontal and Vertical Velocity Component VECTORS

- $x'(t) = \dfrac{dx}{dt}$ is the rate at which the x-coordinate is changing with respect to t or the velocity of a particle in the horizontal direction.

- $y'(t) = \dfrac{dy}{dt}$ is the rate at which the y-coordinate is changing with respect to t or the velocity of a particle in the vertical direction.

- $\bar{s} = \langle x(t), y(t) \rangle = (x(t), y(t))$ is the position at any time t.

Position

- $\bar{v} = \langle x'(t), y'(t) \rangle = (x'(t), y'(t))$ is the velocity vector at any time t.

- $\bar{a} = \langle x''(t), y''(t) \rangle = (x''(t), y''(t))$ is the acceleration vector at any time t.

  *note: the vectors may or may not be contained within the chevrons $\langle\ \rangle$.

- $\dfrac{dy}{dx}$ is the rate of change of y with respect to x or the slope of the tangent line to the curve or the slope of the position vector.

- $\dfrac{d^2y}{dx^2}$ is the rate of change of the slope of the curve with respect to x.

- $|\bar{v}(t)| = \|\bar{v}(t)\| = \sqrt{\left(\dfrac{dx}{dt}\right)^2 + \left(\dfrac{dy}{dt}\right)^2}$ is the <u>**speed of a particle**</u> or the <u>**magnitude**</u> or the <u>**length**</u> or the <u>**norm**</u> of the velocity vector.

- $\int_a^b \sqrt{\left(\dfrac{dx}{dt}\right)^2 + \left(\dfrac{dy}{dt}\right)^2}\, dt$ is the **length of the arc** from $t = a$ to $t = b$ or the **distance traveled** by a particle from $t = a$ to $t = b$.

  *Remember that $\int_a^b |v(t)|\, dt$ is the total distance traveled, whether it be along a straight line or curve.

$\vec{s}(1)=\langle 5,0\rangle$

Example 8:

(No Calculator) A particle moves in the xy-plane so that at any time t, $t \geq 0$, the position of the particle is given by $x(t)=t^3+4t^2$, $y(t)=t^4-t^3$.

VECTOR

$\vec{s}(t)=\langle t^3+4t^2, t^4-t^3\rangle$

(a) Find the velocity vector at $t=1$, (b) the speed of the particle at $t=1$, and (c) the acceleration vector at $t=1$.

a) $\vec{v}(t)=\langle 3t^2+8t, 4t^3-3t^2\rangle$

$\vec{v}(1)=\langle 11, 1\rangle$

MOVING RIGHT 11
UP 1

c) $\vec{a}(t)=\langle 6t+8, 12t^2-6t\rangle$

$\vec{a}(1)=\langle 14, 6\rangle$

b) $\|\vec{v}(1)\|=\sqrt{11^2+1^2}$

$=\sqrt{122}$

Example 9:

(No Calculator) A particle moves in the xy-plane so that $x=\sqrt{3}-4\cos t$ and $y=1-2\sin t$, where $0 \leq t \leq 2\pi$

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

$$\frac{dy}{dx}=\frac{-2\cos t}{4\sin t}=\boxed{-\frac{1}{2}\cot t}$$

$$\frac{d^2y}{dx^2}=\frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}=\frac{+\frac{1}{2}\csc^2 t}{4\sin t}$$

$$=\frac{1}{8}\csc^3 t$$

(b) The path of the particle intersects the x-axis twice. Write an expression that represents the distance traveled by the particle between the two x-intercepts. Do not evaluate.

$1-2\sin t=0$

$\sin t=\frac{1}{2}$

$t=\frac{\pi}{6}, \frac{5\pi}{6}$

$$L=\int_{\pi/6}^{5\pi/6}\sqrt{(4\sin t)^2+(-2\cos t)^2}\,dt$$

$$=\int_{\pi/6}^{5\pi/6}\sqrt{16\sin^2 t+4\cos^2 t}\,dt$$

① $x = t^2 - 1,\ y = e^{t^3}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2 e^{t^3}}{2t} = \boxed{\frac{3}{2} t e^{t^3}}$$

③ $x = t^5 - 1,\ y = 3t^4 - 2t^3$

$$\vec{s}(t) = \langle t^5 - 1,\ 3t^4 - 2t^3 \rangle$$

$$\vec{s}\,'(t) = \vec{v}(t) = \langle 5t^4,\ 12t^3 - 6t^2 \rangle$$

$$\vec{s}\,''(t) = \vec{v}\,'(t) = \vec{a}(t) = \langle 20t^3,\ 36t^2 - 12t \rangle$$

$$\boxed{\vec{a}(1) = \langle 20, 24 \rangle}$$

⑤ $x'(t) = t + 1,\ t \geq 0,\ y = \ln x$

$$x(t) = \tfrac{1}{2}t^2 + t + C$$

<u>when $t = 0;\ x = 1$:</u>

$$1 = 0 + 0 + C,\ C = 1$$

so $x(t) = \tfrac{1}{2}t^2 + t + 1$

$$x(1) = \tfrac{1}{2} + 1 + 1 = \tfrac{5}{2}$$

when $x = \frac{5}{2}$, $y = \ln\frac{5}{2}$

So at $t = 2$, the particle is at $\boxed{(x, y) = \left(\tfrac{5}{2}, \ln\tfrac{5}{2}\right)}$

⑦ $xy = 10,\ x = 2,\ \frac{dy}{dt} = 3,\ \frac{dx}{dt} = ?$

$$y = 10x^{-1}$$

$$\frac{dy}{dx} = \frac{-10}{x^2}$$

<u>when $x = 2$:</u>

$$\frac{dy}{dx} = \frac{-10}{4} = \frac{dy/dt}{dx/dt}$$

So $-\frac{5}{2} = \frac{3}{dx/dt}$

$$\frac{dx}{dt} = 3\left(-\frac{2}{5}\right) = \boxed{\frac{-6}{5}}$$

② $\vec{s}(t) = \langle \ln(t^2 + 5t),\ 3t^2 \rangle,\ t > 0$

$$\vec{s}\,'(t) = \vec{v}(t) = \left\langle \frac{2t + 5}{t^2 + 5t},\ 6t \right\rangle$$

$$\vec{v}(2) = \left\langle \frac{2(2) + 5}{2^2 + 5(2)},\ 6(2) \right\rangle = \boxed{\left\langle \frac{9}{14},\ 12 \right\rangle}$$

④ $\vec{s}(t) = \left\langle \sin\left(3t - \frac{\pi}{2}\right),\ 3t^2 \right\rangle$

$$\vec{s}\,'(t) = \vec{v}(t) = \left\langle 3\cos\left(3t - \tfrac{\pi}{2}\right),\ 6t \right\rangle$$

$$\vec{v}\left(\tfrac{\pi}{2}\right) = \left\langle 3\cos(\pi),\ \tfrac{6\pi}{2} \right\rangle = \boxed{\langle -3,\ 3\pi \rangle}$$

⑥ $\vec{v}(t) = \langle 1 + t,\ t^3 \rangle,\ \vec{s}(0) = \langle 5, 0 \rangle = \langle x(0), y(0) \rangle$

$$x(2) = 5 + \int_0^2 (1 + t)\,dt = 5 + \left[t + \tfrac{1}{2}t^2\right]_0^2$$

$$= 5 + [(2 + 2) - 0] = \boxed{9}$$

$$y(2) = 0 + \int_0^2 t^3\,dt = \tfrac{1}{4}t^4 \Big|_0^2 = \tfrac{16}{4} - 0 = \boxed{4}$$

So $\boxed{\vec{s}(2) = \langle 9, 4 \rangle}$ or particle is at $(9, 4)$ at $t = 2$.

⑧ $x(t) = t^3 - \frac{3}{2}t^2 - 18t + 5,\ y(t) = t^3 - 6t^2 + 9t + 4$

particle is at rest when $\vec{v}(t) = \langle x'(t), y'(t) \rangle = \langle 0, 0 \rangle$

$$x'(t) = 3t^2 - 3t - 18 = 0$$

$$3(t^2 - t - 6) = 0$$

$$3(t - 3)(t + 2) = 0$$

$$\underline{t = 3},\ t = -2$$

$$y'(t) = 3t^2 - 12t + 9 = 0$$

$$3(t^2 - 4t + 3) = 0$$

$$3(t - 3)(t - 1) = 0$$

$$\underline{t = 3},\ t = 1$$

When $\boxed{t = 3}$, $\vec{v}(3) = \langle 0, 0 \rangle$

So the particle is not moving in either the horizontal or vertical direction and is, thus, at rest.

⑨ $x(t)=t^3,\ y(t)=t^2-5t+2$
tangent line at $(x,y)=(8,-4)$
$x'(t)=\frac{dx}{dt}=3t^2,\ y'(t)=\frac{dy}{dt}=2t-5$

Find t @ $(8,-4)$:
$8=t^3$ so $\boxed{t=2}$
$-4=t^2-5t+2$
$t^2-5t+6=0$
$(t-3)(t-2)=0$
$t=3,\ \boxed{t=2}$

so $\left.\frac{dy}{dx}\right|_{t=2}=\frac{dy/dt|_{t=2}}{dx/dt|_{t=2}}$

$=\frac{y'(2)}{x'(2)}=\frac{-1}{12}=m$

so tangent line eq. is

$$\boxed{y=-4-\tfrac{1}{12}(x-8)}$$

⑫ $\frac{dx}{dt}=\frac{1}{t+1},\ \frac{dy}{dt}=2t,\ t\geq 0$

(a) $t=1,\ x=\ln 2,\ y=0$
$\int\frac{1}{t+1}dt=\ln|t+1|+C=x(t)$
$x(1)=\ln 2+C=\ln 2$, so $C=0$
$\int 2t\,dt=t^2+C=y(t)$
$y(1)=1+C=0,\ C=-1$
So $P(x,y)=(x(t),y(t))=\boxed{(\ln|t+1|,\ t^2-1)}$

(b) eliminate parameter: $x=\ln|t+1|,\ t\geq 0$
$e^x=t+1,\ t=e^x-1$
$y=t^2-1\rightarrow\boxed{y=(e^x-1)^2-1}$

(c) Avg R-O-C $=\frac{y(4)-y(0)}{x(4)-x(0)}=\frac{15-(-1)}{\ln 5-\ln 1}=\boxed{\frac{16}{\ln 5}}$

(d) $\frac{dy}{dx}=2(e^x-1)'\cdot e^x$
$\left.\frac{dy}{dx}\right|_{\substack{t=1\\x=\ln 2}}=2e^{\ln 2}(e^{\ln 2}-1)$
$=2(2)(2-1)$
$=\boxed{4}$

⑩ $x(t)=5t+3\sin t,\ y(t)=(8-t)(1-\cos t)$
Find $\vec{v}(t)=\langle x'(t),y'(t)\rangle$ when $x=25$.
$25=5t+3\sin t$
$t=5.445\ldots=A$ (find and store on calculator)

So $\vec{v}(A)=\langle x'(A),y'(A)\rangle$
$$\boxed{\vec{v}(A)=\langle 7.008,\ -2.228\rangle}$$
MATH #8 on calculator at $t=\boxed{A}$

⑪ $t\geq 0,\ x(t)=t^2-3,\ y(t)=\frac{2}{3}t^3$

(a) $\|\vec{v}(5)\|=?$, $x'(t)=2t$, $y'(t)=2t^2$
$x'(5)=10$, $y'(5)=50$
$\|\vec{v}(5)\|=\sqrt{10^2+50^2}=\sqrt{2600}=\boxed{10\sqrt{26}}$

(b) Total Distance = Arc Length $=\int_a^b|v(t)|\,dt$
$=\int_0^5\sqrt{(x'(t))^2+(y'(t))^2}\,dt$
$=\int_0^5\sqrt{4t^2+4t^4}\,dt=\int_0^5\sqrt{4t^2(1+t^2)}\,dt$
$=\int_0^5 2t(1+t^2)^{1/2}dt=\frac{2}{3}(1+t^2)^{3/2}\Big|_0^5$
$=\boxed{\frac{2}{3}\left[26^{3/2}-1^{3/2}\right]}$

(c) $y=\frac{2}{3}t^3\rightarrow t=\left(\frac{3}{2}y\right)^{1/3}$ *eliminate the parameter
so $x=\left(\left(\frac{3}{2}y\right)^{1/3}\right)^2-3=\left(\frac{3}{2}y\right)^{2/3}-3$
*Now solve for y
$x=\left(\frac{3}{2}\right)^{2/3}y^{2/3}-3$
$\left(\frac{3}{2}\right)^{-2/3}(x+3)=y^{2/3}$
$y=\left(\left(\frac{3}{2}\right)^{-2/3}(x+3)\right)^{3/2}$
$y=\frac{2}{3}(x+3)^{3/2}$
$$\boxed{\frac{dy}{dx}=(x+3)^{1/2}}$$

(13) $x = 2 - 3\cos t,\ y = 3 + 2\sin t,\ t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

(a) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \boxed{\frac{2\cos t}{3\sin t}}$ or $= \boxed{\frac{2}{3}\cot t}$

(b) $\left.\frac{dy}{dx}\right|_{t=\pi/4} = \frac{2\cos(\frac{\pi}{4})}{3\sin(\frac{\pi}{4})} = \frac{2(\frac{\sqrt{2}}{2})}{3(\frac{\sqrt{2}}{2})} = \boxed{\frac{2}{3}} = m$

$x(\frac{\pi}{4}) = 2 - 3\cos\frac{\pi}{4} = 2 - \frac{3\sqrt{2}}{2}$

$y(\frac{\pi}{4}) = 3 + 2\sin(\frac{\pi}{4}) = 3 + \sqrt{2}$

tangent line eq: $\boxed{y = (3+\sqrt{2}) + \frac{2}{3}\left(x - \left(2 - \frac{3\sqrt{2}}{2}\right)\right)}$

(c) On y-axis, $x = 0$

$2 - 3\cos t = 0$

From calculator on $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$t = -0.841 = A$

$t = 0.841 = B$

Arclength

$= \int_A^B \sqrt{(2-3\cos t)^2 + (3+2\sin t)^2}\, dt$

$= \boxed{5.196}$ (from calculator)

$y_1 = 2 - 3\cos t$

$y_2 = 3 + 2\sin t$

(14) $x = \sin t,\ y = \csc t,\ 0 < t < \frac{\pi}{2}$

$x'(t) = \cos t = \frac{dx}{dt}$

$x''(t) = -\sin t$

$y'(t) = -\csc t \cot t = -\frac{\cos t}{\sin^2 t}$

$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{-\csc t \cot t}{\cos t}$

$= -\frac{1}{\sin^2 t} < 0$

for $0 < t < \frac{\pi}{2}$

So Decreasing

$\frac{dy}{dx} = -(\csc t)^2,\quad \frac{d^2y}{dx^2} = \frac{-2\csc t \cdot (-\csc t \cot t)}{\cos t}$

$= \frac{2\csc^2 t \cot t}{\cos t} = \frac{2}{\sin^3 t} > 0$

for $0 < t < \frac{\pi}{2}$, so

Concave up $\boxed{C}$

(15) $x = \ln t,\ y = t,\ t > 0$

*eliminate the parameter

$x = \ln y,\ \boxed{y = e^x},\ y > 0$ on $e^x\ \forall x$

$y > 0$

$\boxed{C}$

(16) $x(t) = t^2 + 1,\ y = \ln(2t+3),\ t \geq 0$

$x'(t) = 2t,\quad y'(t) = \frac{2}{2t+3} = 2(2t+3)^{-1}$

$x''(t) = 2,\quad y''(t) = -2(2t+3)^{-2}(2) = \frac{-4}{(2t+3)^2}$

so $\vec{a}(t) = \left\langle 2, -\frac{4}{(2t+3)^2} \right\rangle$

or $\left(2, -\frac{4}{(2t+3)^2}\right)$ $\boxed{E}$

*vectors aren't always written in chevrons < >.

§9.2—Parametric & Vector Accumulation

$\int_a^1 v(t)\,dt$ $\int_a^b |v(t)|\,dt$

Displacement & Distance Traveled

Suppose a particle moves along a path in the plane so that its velocity at any time t is $\vec{v}(t) = (x'(t), y'(t))$, then the **displacement** from $t = a$ to $t = b$ is given by the vector

$$\left\langle \int_a^b x'(t)\,dt, \int_a^b y'(t)\,dt \right\rangle.$$

The preceding vector is added to the position at time $t = a$ to get the **position** at time $t = b$.

The **distance traveled** from $t = a$ to $t = b$ is the arc length

$$\int_a^b |\vec{v}(t)|\,dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2}\,dt.$$

Example 1:
(Calculator) An object moving along a curve in the xy-plane has position $(x(t), y(t))$ at time t with $\frac{dx}{dt} = t\sin(t)$, $\frac{dy}{dt} = \cos(t^2)$. At time $t = 2$, the object is at the position $(1, 4)$.

a) Find the acceleration vector for the object at $t = 2$.

$\vec{a}(2) = \langle x''(2), y''(2) \rangle$

$= \langle .077, 3.027 \rangle = \text{IF } \langle A, B \rangle$

b) Write the equation of the tangent line to the curve at the point where $t = 2$.

$(1,4)$

$m = \frac{y'(2)}{x'(2)} = \frac{\cos 4}{2\sin 2}$ EQ: $y = 4 + \frac{\cos 4}{2\sin 2}(x-1)$

$y = 4 - \cancel{0.297}\,(x-1)$
0.359

OR NORMAL OR LINE

c) Find the speed of the object at $t = 2$.

$\|\vec{v}(2)\| = \sqrt{(x'(2))^2 + (y'(2))^2}$

$= \sqrt{(2\sin 2)^2 + (\cos 4)^2}$

$= 1.932$

d) Find the displacement of the object from $t=2$ to $t=7$.

DISPLACEMENT OF x: $\int_2^7 x'(t)dt = -6.362 \to A$

DISPLACEMENT OF y: $\int_2^7 y'(t)dt = 0.097 \to B$

e) Find the distance traveled by the object from $t=2$ to $t=7$.

$L = \int_2^7 \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$

$L = 13.229$

f) Find the position of the object at time $t=1$.

$(x(1), y(1)) = \left(1 + \int_2^1 x'(t)dt,\ 4 + \int_2^1 y'(t)dt\right)$

$= (-.440, 4.443)$

Example 2: CALCULATOR

An object moving along a curve in the xy-plane has position $(x(t), y(t))$ at time $t \geq 0$ with $\frac{dx}{dt} = 2\sin(t^2)$. The derivative $\frac{dy}{dt}$ is not explicitly given. At time $t=2$, the object is at position $(3,5)$.

a) Find the x-coordinate of the position of the object at time $t=4$.

$x(4) = 3 + \int_2^4 x'(t)dt$

$= 2.885$

b) At time $t = 2$, the value of $\frac{dy}{dt}$ is -6. Write an equation for the line tangent to the curve at the point $(x(2), y(2))$.

$y'(2) = -6$

Pt: $(3, 5)$

$m = \frac{y'(2)}{x'(2)} = \frac{-6}{2\sin 4}$

EQ: $\boxed{y = 5 - \frac{6}{2\sin 4}(x-3)}$

$y = 5 + 3.964(x-3)$

c) Find the magnitude of the velocity vector at time $t = 2$.

SPEED

$$\|\vec{v}(2)\| = \sqrt{(x'(2))^2 + (y'(2))^2}$$

$$= \sqrt{(2\sin 4)^2 + (-6)^2}$$

d) For $t \geq 3$, the line tangent to the curve at $(x(t), y(t))$ has a slope of $2t - 1$. Find the acceleration vector of the object at time $t = 4$.

$\hookrightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = 2t - 1$

$$\frac{\frac{dy}{dt}}{2\sin(t^2)} = 2t - 1$$

$$y'(t) = \frac{dy}{dx} = (2t-1)(2\sin(t^2))$$

$$\vec{a}(4) = \langle x''(4), y''(4) \rangle$$

$$= \langle -15.322, -108.408 \rangle$$

① $x = e^{2t}$, $y = \sin(3t)$

$\frac{dx}{dt} = 2e^{2t}$, $\frac{dy}{dt} = 3\cos 3t$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \boxed{\frac{3\cos(3t)}{2e^{2t}}}$

② $x = \cos^3 t$, $y = \sin^2 t$, $0 \le t \le \frac{\pi}{2}$

$\text{Length} = \int_0^{\pi/2} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$

$x'(t) = 3\cos^2 t(-\sin t)$, $y'(t) = 2\sin t\cos t \overset{\text{or (Double angle !!)}}{=} 2\sin 2t$

$= -3\sin t\cos^2 t$

so $\boxed{L = \int_0^{\pi/2} \sqrt{9\sin^2 t\cos^4 t + 4\sin^2 t\cos^2 t}\, dt}$

③ $x = t^3 - t^2 - 1$, $y = t^4 + 2t^2 - 8t$

$\frac{dx}{dt} = 3t^2 - 2t$, $\frac{dy}{dt} = 4t^3 + 4t - 8$

vertical tangent when $\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$

$\frac{dy}{dx} = \frac{\neq 0}{0}$

so $3t^2 - 2t = 0$

$t(3t - 2) = 0$

$t = 0$, $t = \frac{2}{3}$

$\left.\frac{dy}{dt}\right|_{t=0} = 8 \neq 0$

$\left.\frac{dy}{dt}\right|_{t=\frac{2}{3}} = \frac{32}{27} + \frac{8}{3} - 8 \neq 0$

So the curve has vertical tangents for $t = 0$ and $t = \frac{2}{3}$

④ $x(t) = 3t^2 - 4t + 2$, $y(t) = t^3 - 4t$

$x'(t) = 6t - 4$, $y'(t) = 3t^2 - 4$

$\left.\frac{dy}{dx}\right|_{t=1} = \frac{3(1^2) - 4}{6(1) - 4} = \frac{-1}{2} = m$

$(x(1), y(1)) = (3 - 4 + 2, 1 - 4) = (1, -3)$

so tangent line eq. is

$\boxed{y = -3 - \frac{1}{2}(x - 1)}$

⑤ $x(t) = e^t + 1$, $y = 2e^{2t}$ *eliminate the parameter

$e^t = x - 1$

$t = \ln(x-1)$ plug in

$y = 2e^{2\ln(x-1)}$

$y = 2e^{\ln(x-1)^2}$

$\boxed{y = 2(x-1)^2}$

⑥ $x = \cos(5t)$, $y = t^3$.

$x'(t) = -5\sin 5t$, $y'(t) = 3t^2$

$\vec{v}(t) = \langle -5\sin 5t, 3t^2 \rangle$

$\vec{v}(2) = \langle -5\sin 10, 12 \rangle$

Speed @ $t = 2$ is $\|\vec{v}(2)\| = \sqrt{25\sin^2 10 + 144}$

$\boxed{\approx 12.304}$

⑦ $x(t) = \frac{1}{3}(t-2)^3 + 4$, $y(t) = t^2 - 4t + 4$

$\vec{v}(t) = \langle (t-2)^2, 2t - 4 \rangle$

(a) $\vec{v}(1) = \langle 1, -2 \rangle$

$\|\vec{v}(1)\| = \sqrt{1 + 4} = \boxed{\sqrt{5}} \approx 2.236$

(b) $\text{Dist} = \int_0^1 \sqrt{(t-2)^4 + (2t-4)^2}\, dt$

$\approx \boxed{3.815 \text{ or } 3.816}$

(c) Particle at rest when $\vec{v}(t) = 0$, or when

$x'(t) = (t-2)^2 = 0$ & $y'(t) = 2t - 4 = 0$

$t = 2$

$2t = 4$

$t = 2$

So particle is at rest when $\boxed{t = 2}$

⑧ $t \geq 0$, $\frac{dx}{dt} = x'(t) = 1 + \tan(t^2)$, $\frac{dy}{dt} = y'(t) = 3e^{\sqrt{t}}$

$\vec{v}(t) = \langle 1 + \tan(t^2), 3e^{\sqrt{t}} \rangle$

$\boxed{\vec{a}(t) = \vec{v}'(t) = \langle 2t\sec^2(t^2), \frac{3}{2\sqrt{t}} e^{\sqrt{t}} \rangle}$

$\vec{v}(5) = \langle 1 + \tan(25), 3e^{\sqrt{5}} \rangle$

speed at $t=5$ is $\|\vec{v}(5)\| = \sqrt{(1+\tan 25)^2 + (3e^{\sqrt{5}})^2}$

$= \boxed{28.082 \text{ or } 28.083}$

⑨ $x(t) = t + \cos t$, $y(t) = 3t + 2\sin t$, $0 \leq t \leq \pi$

$y = 3t + 2\sin t = 5$

when $t = 1.079... = A$ ← store as A

From calculator

$\vec{v}(A) = \langle x'(A), y'(A) \rangle$

$= \boxed{\langle 0.1185, 3.944 \rangle}$

⑩ $\frac{dx}{dt} = x'(t) = 2\sin(t^3)$, $\frac{dy}{dt} = y'(t) = \cos(t^2)$, $0 \leq t \leq 4$

at $t = 1$, $(x(1), y(1)) = (3, 4)$

(a) tangent line at (3,4).

$\left.\frac{dy}{dx}\right|_{(3,4)} = \frac{y'(1)}{x'(1)} = \frac{\cos 1}{2\sin 1} = 0.321 = A$

eq: $\boxed{y = 4 + 0.321(x - 3)}$

(b) speed at $t=2$ is $\|\vec{v}(2)\|$

$= \sqrt{(2\sin 8)^2 + (\cos 4)^2} = \boxed{2.083} = B$

(c) Dist $= \int_0^1 \sqrt{(2\sin(t^3))^2 + (\cos(t^2))^2}\, dt$

$= \boxed{1.126}$

(d) Position at $t = 2$

$(x(2), y(2)) = \left(3 + \int_1^2 x'(t)\,dt,\ 4 + \int_1^2 y'(t)\,dt\right)$

$= \boxed{(3.436, 3.556)}$

⑪ $\frac{dx}{dt} = \cos(t^2)$, $\frac{dy}{dt} = \sin(t^3)$, at $t = 0$, $(x(0), y(0)) = (4, 7)$

$(x(2), y(2)) = \left(4 + \int_0^2 \cos(t^2)\,dt,\ 7 + \int_0^2 \sin(t^3)\,dt\right) = (4.461, 7.451)$

or $\boxed{\langle 4.461, 7.452 \rangle}$ $\boxed{D}$

(12) $x = \sin(2t)$, $y = \cos(5t)$

speed at $t=2$ is $\sqrt{(2\cos(2\cdot 2))^2 + (-5\sin(5\cdot 2))^2} \approx \boxed{3.0179..}$ $\boxed{B}$

(13) $x(t) = t^3 - t^2 - 1$, $y(t) = t^4 + 2t^2 - 8t$

$x'(t) = 3t^2 - 2t$, $y'(t) = 4t^3 + 4t - 8$

vert tangent when $\frac{dy}{dx} = \frac{\neq 0}{0}$, or when $x'(t) = 0$, $y'(t) \neq 0$

$3t^2 - 2t = 0$

$t(3t-2) = 0$

when $t = 0$, $t = \frac{2}{3}$

$y'(0) = -8 \neq 0$

$y'(\frac{2}{3}) = 4(\frac{8}{27}) + 4(\frac{2}{3}) - 8 \neq 0$

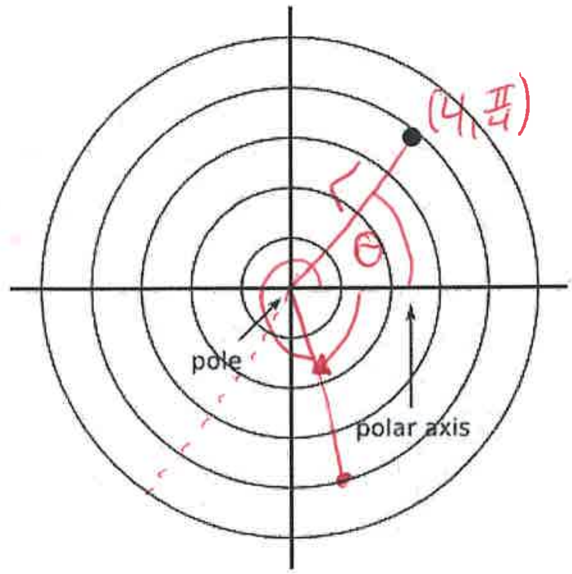
So the curve has vertical tangents at $\boxed{t=0 \text{ and } t=\frac{2}{3}}$ $\boxed{C}$

(14) $\vec{s}(t) = \langle t^2, t \rangle$, $0 \leq t \leq 4$, $\vec{s}'(t) = \vec{v}(t) = \langle 2t, 1 \rangle$

$\boxed{\text{Dist} = \int_0^4 \sqrt{4t^2 + 1}\, dt}$ $\boxed{D}$

ANSWER KEY

§10.1—Polar Intro & Derivatives



A rectangular coordinate system is only one way to navigate through a Euclidean plane. Such coordinates, (x, y), known as **rectangular coordinates**, are useful for expressing functions of y in terms of x. Curves that are not functions are often more easily expressing in an alternative coordinate system called **polar coordinates**.

In a polar coordinate system, we still have the traditional x- and y- axes. The intersection of these axes, the old origin, is called the **pole**. Similar to navigating on the Unit Circle, we can now get to any point in 2-D space by specifying an independent choice of an angle, θ, from the initial ray, **polar axis**, then walking out along that terminal ray a specified amount, r, in either direction.

Although the angle is the independent variable, we express the point in the polar plane as (r, θ). The point to the right would have coordinates of $\left(4, \frac{\pi}{4}\right)$.

Example 1:
Find several other equivalent polar coordinates for the point shown above, then find the equivalent rectangular coordinate.

$\left(-4, \frac{5\pi}{4}\right), \left(-4, -\frac{3\pi}{4}\right), \left(4, -\frac{7\pi}{4}\right)$

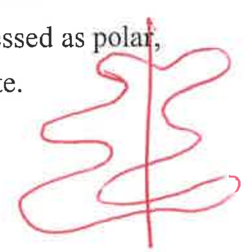
$\sin\theta = \frac{y}{r} \rightarrow y = r\sin\theta$

$\cos\theta = \frac{x}{r} \rightarrow x = r\cos\theta$

$(x, y) = \left(4\cos\frac{\pi}{4}, 4\sin\frac{\pi}{4}\right)$

$= (2\sqrt{2}, 2\sqrt{2})$

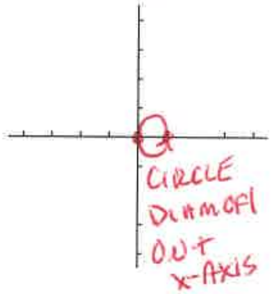
Why use polar coordinates? Graphs that aren't functions in rectangular form $f(x)$ can still be functions in polar form $r(\theta)$. Some of these curves can be quite elaborate and are more easily expressed as polar, rather than rectangular equations, as the following calculator exploration will demonstrate.



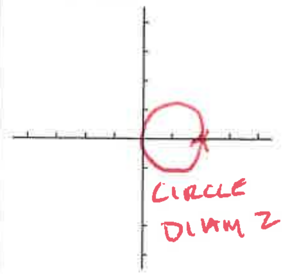
Put your graphing calculator in POLAR mode and RADIAN mode. Graph the following equations on your calculator, sketch the graphs on this sheet, and answer the questions.

Example 2:

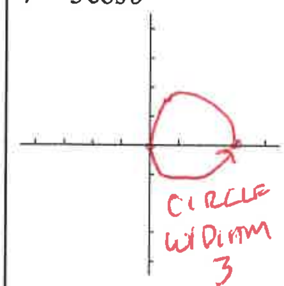
$r = \cos\theta$ $\to r(\theta) = \cos\theta$



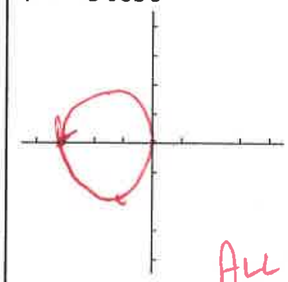
$r = 2\cos\theta$



$r = 3\cos\theta$

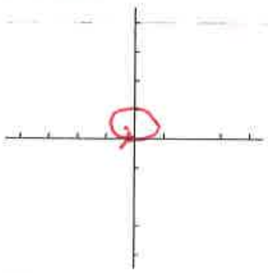


$r = -3\cos\theta$

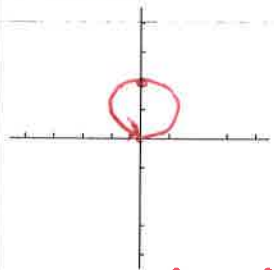


ALL COUNTER CLOCKWISE

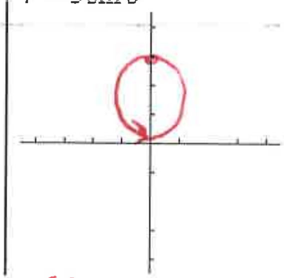
$r = \sin\theta$



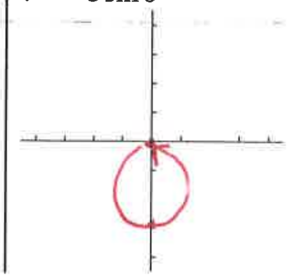
$r = 2\sin\theta$



$r = 3\sin\theta$



$r = -3\sin\theta$

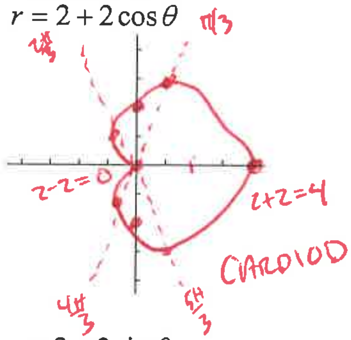


What do you notice about these graphs? ALL CIRCLES
DIAM $= a$
CCW $[0, \pi]$

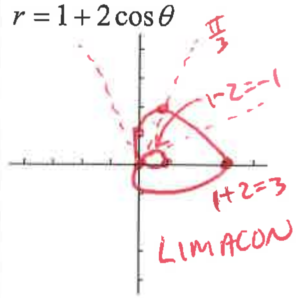
$y = a\cos\theta$
$y = a\sin\theta$

Example 3:

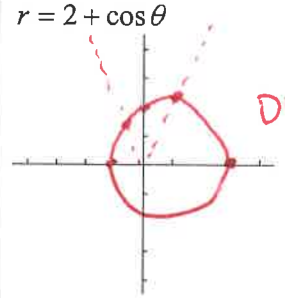
$r = 2 + 2\cos\theta$



$r = 1 + 2\cos\theta$

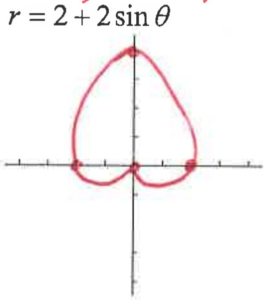


$r = 2 + \cos\theta$

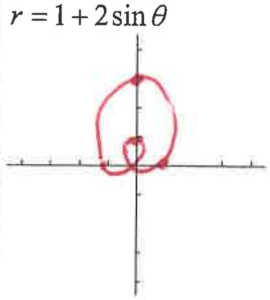


DIMPLED LIMACON

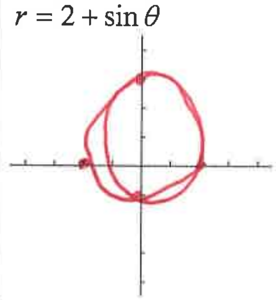
$r = 2 + 2\sin\theta$



$r = 1 + 2\sin\theta$



$r = 2 + \sin\theta$



Which graphs go through the pole?
Which ones do not go through the pole?
Which ones have an inner loop?
What causes these things to happen? (Hint: Go to FORMAT and set your calculator to see the Polar Graphing Coordinates when you trace.)

cos x → x Axis Sym
sin y → y Axis Sym

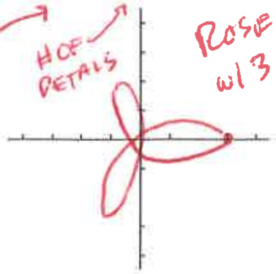
Example 4:

$r = 3\cos 3\theta$

PETAL LENGTH

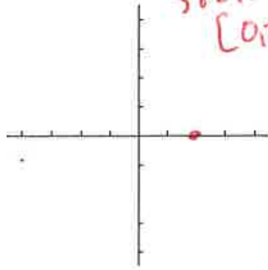
OF PETALS

ROSE CURVE w/ 3 PETAL [0, π]



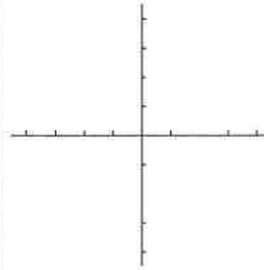
$r = 2\cos 5\theta$

5 PETALS [0, π]

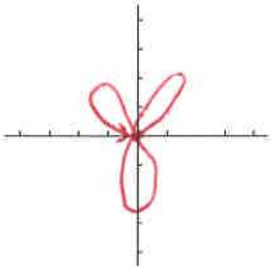


$r = 4\cos 7\theta$

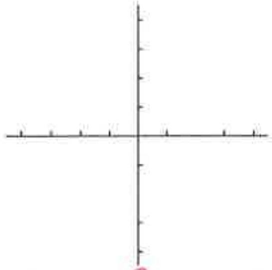
7 PETALS [0, π]



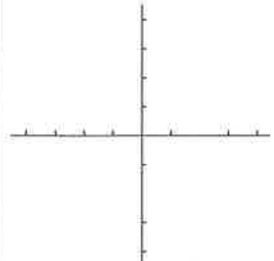
$r = 3\sin 3\theta$



$r = 2\sin 5\theta$



$r = 4\sin 7\theta$



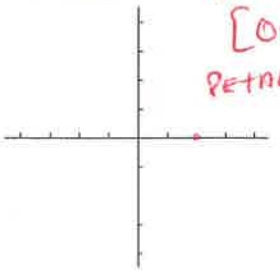
What do you notice about these graphs? ALL POLAR, ODD # OF PETALS, [0, π]

Example 5:

$r = 3\cos 2\theta$

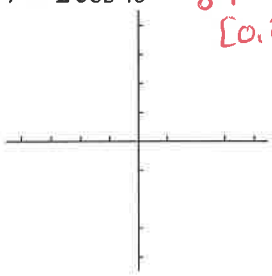
4 PETALS [0, 2π]

PETAL LENGTH 3



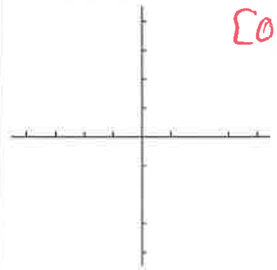
$r = 2\cos 4\theta$

8 PETALS [0, 2π]



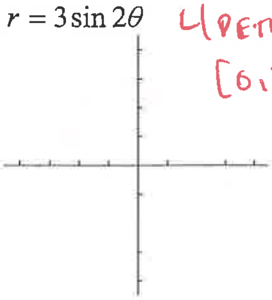
$r = 4\cos 6\theta$

12 PETALS [0, 2π]



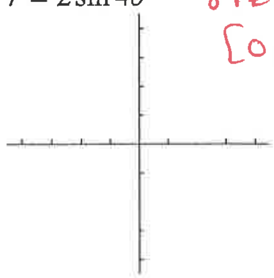
$r = 3\sin 2\theta$

4 PETALS [0, 2π]



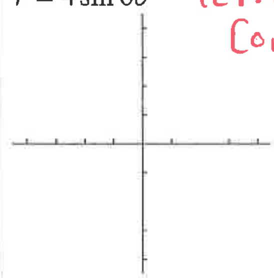
$r = 2\sin 4\theta$

8 PETALS [0, 2π]



$r = 4\sin 6\theta$

12 PETALS [0, 2π]



What do you notice about these graphs?

Before we jump into the calculus of polar, we need to develop some proficiency converting back and forth from polar to rectangular coordinates and equations.

Rectangular/Polar Coordinates and Equations

Rectangular coordinates are in the form (x, y), where x is the independent variable.

Polar coordinates are in the form (r, θ), where θ is the independent variable.

For coordinate conversions:

Polar to Rectangular

$x = r\cos\theta$

$y = r\sin\theta$

$(r, \theta) \to (x, y)$

Rectangular to Polar

$r = \sqrt{x^2 + y^2}$

$\tan\theta = \dfrac{y}{x}$

Example 6:
Convert the following coordinates as specified.

a) Convert $\left(2, \dfrac{5\pi}{6}\right)$ to rectangular coordinates.

(r, θ)

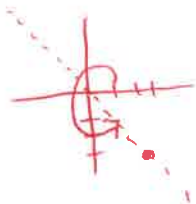
$(-2, -\frac{\pi}{6})$

$(-2, \frac{11\pi}{6})$

$(x, y) = (2\cos\frac{5\pi}{6}, 2\sin\frac{5\pi}{6})$

$= \boxed{(-\sqrt{3}, 1)}$

b) Convert $(3, -3)$ to polar coordinates.



$\tan\theta = \frac{-3}{3} = -1$

$(3\sqrt{2}, -\frac{\pi}{4})$

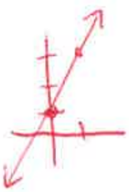
$(3\sqrt{2}, \frac{7\pi}{4})$

$(-3\sqrt{2}, \frac{3\pi}{4})$

$(-3\sqrt{2}, -\frac{5\pi}{4})$

Example 7:
Convert the following equations from rectangular to polar.

a) $y = 2x + 1$

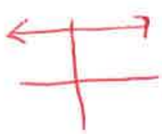


$r\sin\theta = 2r\cos\theta + 1$

$r(\sin\theta - 2\cos\theta) = 1$

$\boxed{r = \dfrac{1}{\sin\theta - 2\cos\theta}}$

b) $y = 4$



$r\sin\theta = 4$

$r = \dfrac{4}{\sin\theta}$

$\boxed{r = 4\csc\theta}$

c) $x^2 + y^2 = 16$ $\to$ $r^2 = 16$, $r = \pm 4$

$(r\cos\theta)^2 + (r\sin\theta)^2 = 16$

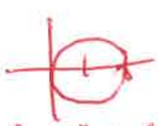
$r^2(\cos^2\theta + \sin^2\theta) = 16$

$r^2 = 16$

$\boxed{r = \pm 4}$

Example 8:
Convert the following equations from polar to rectangular.

a) $r = 2\cos\theta$



$r = 2\left(\frac{x}{r}\right)$

$r^2 = 2x$

$x^2 + y^2 = 2x$

$x^2 - 2x + \underline{(-1)^2} + y^2 = \underline{(-1)^2}$

$x^2 - 2x + 1 + y^2 = 1$

$\boxed{(x-1)^2 + y^2 = 1}$

$(x-h)^2 + (y-k)^2 = r^2$

$\boxed{(x-1)^2 + y^2 = 1}$

b) $\theta = \dfrac{3\pi}{4}$

$\tan\theta = \tan\frac{3\pi}{4}$

$\frac{y}{x} = -1$

$\boxed{y = -x}$

c) $r = \csc\theta$

$r = \frac{r}{y}$

$yr = r$

$\boxed{y = 1}$

Now it's time to B.O.T.C.

Slope of a Polar Equation

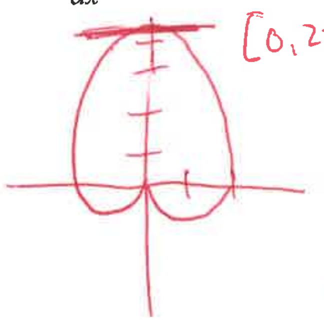
To find the slope of a tangent line to a polar graph $r = f(\theta)$, we can use $x = r\cos\theta$ and $y = r\sin\theta$ and the product rule:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{r\cos\theta + r'\sin\theta}{-r\sin\theta + r'\cos\theta}, \text{ provided that } \frac{dx}{d\theta} \neq 0$$

Or . . . . create $y(\theta)$ and $x(\theta)$ first, then find $y'(\theta)$ and $x'(\theta)$.

Example 9:

Find $\frac{dy}{dx}$ and the slope of the graph of the polar curve at the given value of θ.



$[0, 2\pi]$

$$r = 2 + 2\sin\theta \text{ at } \theta = \frac{\pi}{2}$$

$\frac{dr}{d\theta} = 2\cos\theta$

$$\frac{dy}{dx} = \frac{(2+2\sin\theta)\cos\theta + 2\cos\theta\sin\theta}{-(2+2\sin\theta)\sin\theta + 2\cos\theta\cos\theta} = \frac{2\cos\theta + 2\sin\theta\cos\theta + 2\sin\theta\cos\theta}{-2\sin\theta - 2\sin^2\theta + 2\cos^2\theta}$$

$$= \frac{2\cos\theta + 4\sin\theta\cos\theta}{2(\cos^2\theta - \sin^2\theta - \sin\theta)} = \frac{2(\cos\theta + 2\sin\theta\cos\theta)}{2(1-\sin^2\theta - \sin^2\theta - \sin\theta)} = \frac{2\cos\theta(1+2\sin\theta)}{2(1-\sin\theta - 2\sin^2\theta)}$$

$$= \frac{\cancel{2}\cos\theta(1+2\sin\theta)}{\cancel{2}(1-2\sin\theta)(1+\sin\theta)}\Bigg|_{\theta=\frac{\pi}{2}} \frac{\cos\frac{\pi}{2}(1+2\sin\frac{\pi}{2})}{(1-2\sin\frac{\pi}{2})(1+\sin\frac{\pi}{2})} = \frac{0}{(-1)(2)} = \frac{0}{-2} \to \boxed{0}$$

Example 10:

Find the **points,** (r,θ) of horizontal and vertical tangency to the graph of $r = 2 - 2\cos\theta$. Beware of $\frac{0}{0}$.

$\frac{dy}{d\theta} = 0$ $\qquad$ $\frac{dx}{d\theta} = 0$

$\frac{dr}{d\theta} = 2\sin\theta$

$$\frac{dy}{dx} = \frac{(2-2\cos\theta)\cos\theta + 2\sin\theta(\sin\theta)}{-(2-2\cos\theta)\sin\theta + 2\sin\theta\cos\theta} = \frac{2\cos\theta - 2\cos^2\theta + 2\sin^2\theta}{-2\sin\theta + 2\cos\theta\sin\theta + 2\cos\theta\sin\theta}$$

$1 + x - 2x^2$

$(1-x)(1+2x)$

$$= \frac{\sin^2\theta - \cos^2\theta + \cos\theta}{2\sin\theta\cos\theta - \sin\theta} = \frac{1-\cos^2\theta - \cos^2\theta + \cos\theta}{\sin\theta(2\cos\theta - 1)} = \frac{1+\cos\theta - 2\cos^2\theta}{\sin\theta(2\cos\theta-1)}$$

$$= \cancel{\frac{-\cos 2\theta + \cos\theta}{\sin 2\theta - \sin\theta}} = \frac{(1-\cos\theta)(1+2\cos\theta)}{\sin\theta(2\cos\theta - 1)}$$

HORIZ TANG

$1+\cos\theta = 0$ $\qquad$ $1+2\cos\theta = 0$

$\cos\theta = -1$ $\qquad$ $\cos\theta = -\frac{1}{2}$

$\theta = \pi$ $\qquad$ $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

VERT TANG

$\sin\theta = 0$ $\qquad$ $2\cos\theta - 1 = 0$

$\qquad$ $\cos\theta = \frac{1}{2}$

$\theta = 0, 2\pi$ $\qquad$ $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$

(1) $y = 4$

$y = r\sin\theta$

so $4 = r\sin\theta$

$r = \frac{4}{\sin\theta}$

or $\boxed{r = 4\csc\theta}$

(2) $3x - 5y + 2 = 0$

* $x = r\cos\theta$, $y = r\sin\theta$

so $3(r\cos\theta) - 5(r\sin\theta) + 2 = 0$

$r(3\cos\theta - 5\sin\theta) = -2$

$\boxed{r = \frac{-2}{3\cos\theta - 5\sin\theta}}$

(3) $x^2 + y^2 = 25$

* $x^2 + y^2 = r^2$

So $r^2 = 25$

$\boxed{r = 5 \text{ or } r = -5}$ traces out differently, but same graph

(4) $r = 3\sec\theta$

* $\sec\theta = \frac{r}{x}$

$r = 3\left(\frac{r}{x}\right)$

$1 = \frac{3}{x}$

$\boxed{x = 3}$

(5) $r = 2\sin\theta$

* $\sin\theta = \frac{y}{r}$

$r = 2\left(\frac{y}{r}\right)$

$r^2 = 2y$

$\boxed{x^2 + y^2 = 2y}$

(6) $\theta = \frac{5\pi}{6}$

$\tan\theta = \tan\frac{5\pi}{6}$ * take tangent of both sides

$\frac{y}{x} = -\frac{\sqrt{3}}{3}$

$\boxed{y = -\frac{\sqrt{3}}{3}x}$

(7) $r = 2 + 3\sin\theta$, $\theta = \frac{3\pi}{2}$

$x = r\cos\theta$, $y = r\sin\theta$

$x = (2 + 3\sin\theta)\cos\theta$, $y = (2 + 3\sin\theta)\sin\theta$

$x'(\theta) = (3\cos\theta)(\cos\theta) + (2 + 3\sin\theta)(-\sin\theta)$

$y'(\theta) = (3\cos\theta)(\sin\theta) + (2 + 3\sin\theta)(\cos\theta)$

$x'\left(\frac{3\pi}{2}\right) = 3\left(\cos\frac{3\pi}{2}\right)^2 - \left(\sin\frac{3\pi}{2}\right)\left(2 + 3\sin\frac{3\pi}{2}\right)$

$= 0 - (-1)(2 - 3) = -1$

$y'\left(\frac{3\pi}{2}\right) = \left(3\cos\frac{3\pi}{2}\right)\left(\sin\frac{3\pi}{2}\right) + \left(2 + 3\sin\frac{3\pi}{2}\right)\left(\cos\frac{3\pi}{2}\right)$

$= 0 + (2 - 3)(0) = 0$

$\left.\frac{dy}{dx}\right|_{\theta = \frac{3\pi}{2}} = \frac{y'\left(\frac{3\pi}{2}\right)}{x'\left(\frac{3\pi}{2}\right)} = \frac{0}{-1} = \boxed{0}$

(8) $r = 3(1 - \cos\theta)$, $\theta = \frac{\pi}{2}$

$x = 3(1 - \cos\theta)\cos\theta = 3\cos\theta - 3\cos^2\theta$

$x' = -3\sin\theta - 6\cos\theta(-\sin\theta)$

$x'\left(\frac{\pi}{2}\right) = -3\sin\frac{\pi}{2} + 6\cos\frac{\pi}{2}\sin\frac{\pi}{2}$

$= \boxed{-3}$

$y = 3(1 - \cos\theta)\sin\theta$

$y' = 3(\sin\theta)\sin\theta + 3(1 - \cos\theta)\cos\theta$

$y'\left(\frac{\pi}{2}\right) = 3\left(\sin\frac{\pi}{2}\right)^2 + 3\left(1 - \cos\frac{\pi}{2}\right)\cos\frac{\pi}{2}$

$= 3 + 0 = \boxed{3}$

$\left.\frac{dy}{dx}\right|_{\theta = \frac{\pi}{2}} = \frac{y'\left(\frac{\pi}{2}\right)}{x'\left(\frac{\pi}{2}\right)} = \frac{3}{-3} = \boxed{-1}$

(9) $r = 4\sin\theta$, $\theta = \frac{\pi}{3}$

$x = 4\sin\theta\cos\theta = 2\sin 2\theta$, $x' = 4\cos 2\theta$

$x'\left(\frac{\pi}{3}\right) = 4\left(-\frac{1}{2}\right) = \underline{-2}$

$y = 4\sin^2\theta$, $y' = 8\sin\theta\cos\theta = 4\sin 2\theta$

$y'\left(\frac{\pi}{3}\right) = 4\sin\frac{2\pi}{3} = 4\left(\frac{\sqrt{3}}{2}\right) = \underline{2\sqrt{3}}$

$\left.\frac{dy}{dx}\right|_{\theta = \frac{\pi}{3}} = \frac{2\sqrt{3}}{-2} = \boxed{-\sqrt{3}}$

(10) $r = 2\sin(3\theta)$, $\theta = \frac{\pi}{4}$

$x = 2\sin(3\theta)\cos\theta$, $x' = 6\cos(3\theta)\cos\theta - 2\sin 3\theta\sin\theta$

$x'\left(\frac{\pi}{4}\right) = 6\cos\left(\frac{3\pi}{4}\right)\cos\left(\frac{\pi}{4}\right) - 2\sin\left(\frac{3\pi}{4}\right)\sin\left(\frac{\pi}{4}\right) =$

$= 6\left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = -3 - 1 = \underline{-4}$

$y = 2\sin(3\theta)\sin\theta$, $y' = 6\cos(3\theta)\sin\theta + 2\sin(3\theta)\cos\theta$

$y'\left(\frac{\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = -3 + 1 = \underline{-2}$

$\left.\frac{dy}{dx}\right|_{\theta = \pi/4} = \frac{-2}{-4} = \boxed{\frac{1}{2}}$

(11) $r = 1+\sin\theta$

$x = (1+\sin\theta)\cos\theta$, $x' = \cos^2\theta - (1+\sin\theta)\sin\theta = \cos^2\theta - \sin\theta - \sin^2\theta = \cos 2\theta - \sin\theta$

$y = (1+\sin\theta)\sin\theta = \sin\theta + \sin^2\theta$, $y' = \underline{\cos\theta + 2\sin\theta\cos\theta} = \cos\theta + \sin 2\theta$

← easier to use below

Horizontal tangents when $y' = 0$, $x' \neq 0$:

$\cos\theta + 2\sin\theta\cos\theta = 0$

$\cos\theta(1+2\sin\theta) = 0$

$\cos\theta = 0$ or $\sin\theta = -\frac{1}{2}$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ or $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

$(r,\theta) = (1+\sin\theta, \theta)$

pts: $\left(2, \frac{\pi}{2}\right), \cancel{\left(0, \frac{3\pi}{2}\right)}, \left(\frac{1}{2}, \frac{7\pi}{6}\right), \left(\frac{1}{2}, \frac{11\pi}{6}\right)$

throw out → $\frac{0}{0}$

Vert tangents when $x' = 0$, $y' \neq 0$

$\cos 2\theta - \sin\theta = 0$

$(\cos^2\theta) - \sin^2\theta - \sin\theta = 0$

$(1-\sin^2\theta) - \sin^2\theta - \sin\theta = 0$

$-2\sin^2\theta - \sin\theta + 1 = 0$

$2\sin^2\theta + \sin\theta - 1 = 0$

$(2\sin\theta - 1)(\sin\theta + 1) = 0$

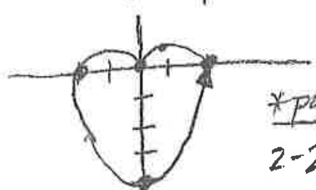
$\sin\theta = \frac{1}{2}$ or $\sin\theta = -1$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ or $\cancel{\theta = \frac{3\pi}{2}}$ ← throw out → $\frac{0}{0}$

pts: $\left(\frac{3}{2}, \frac{\pi}{6}\right), \left(\frac{3}{2}, \frac{5\pi}{6}\right)$

(12) $r = 2 - 2\sin\theta$

θ	0	$\pi/6$	$\pi/2$	π	$3\pi/2$	2π
r	2	1	0	2	4	0



CARDIOID

*polar zeros:

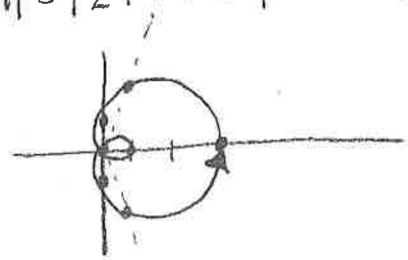
$2 - 2\sin\theta = 0$

$\sin\theta = 1$

$\theta = \frac{\pi}{2}$

(13) $r = 1 + 2\cos\theta$

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π	$4\pi/3$	$3\pi/2$	2π
r	3	2	1	0	-1	0	1	3



LIMAÇON

*polar zeros:

$1 + 2\cos\theta = 0$

$\cos\theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

(14) $r = 4\cos(2\theta)$

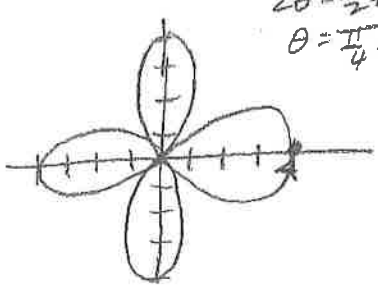
θ	0	$\pi/4$	$\frac{\pi}{2}$
r	4	0	-4

*polar zeros: $4\cos 2\theta = 0$

$\cos 2\theta = 0$

$2\theta = \frac{\pi}{2} + \pi n$

$\theta = \frac{\pi}{4} + \frac{\pi}{2} n$



ROSE CURVE

(15) $r^2 = 4\sin(2\theta)$, $r = \pm\sqrt{4\sin(2\theta)}$

θ	0	$\pi/2$	$3\pi/4$
r	2	0	$\sqrt{-4}$ undefined

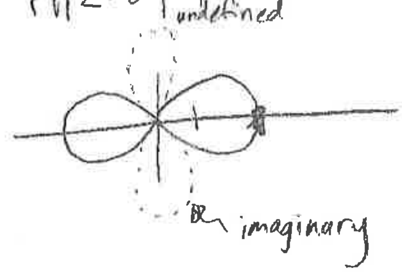
polar zeros:

$4\sin 2\theta = 0$

$\sin 2\theta = 0$

$2\theta = 0 + \pi n$

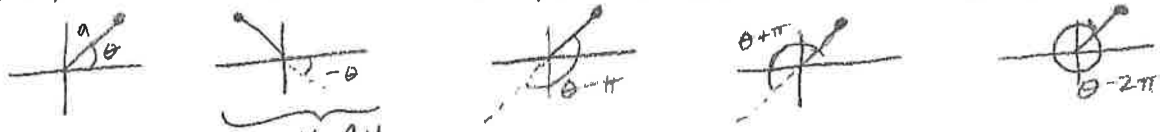
$\theta = 0 + \frac{\pi}{2} n$



LEMNISCATE

(16) $a \neq 0, \theta \neq 0$

(A) (a, θ) (B) $(-a, -\theta)$ (C) $(-a, \theta - \pi)$ (D) $(-a, \theta + \pi)$ (E) $(a, \theta - 2\pi)$



doesn't fit other 4 $\boxed{B}$

(17) $r = f(\theta)$

$y = r\sin\theta$, Slope $= \dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$ $\boxed{D}$

$x = r\cos\theta$

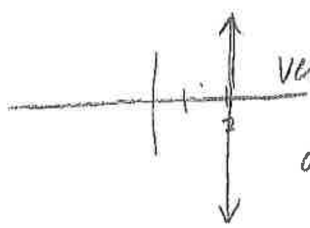
(18) $r = 2\sec\theta$

$r = 2\left(\frac{r}{x}\right)$

$(r) = \frac{2}{x}(r)$

$1 = \frac{2}{x}$

$x = 2$



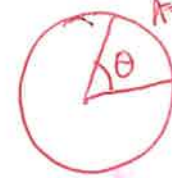
vertical line at $x = 2$ $\boxed{D}$

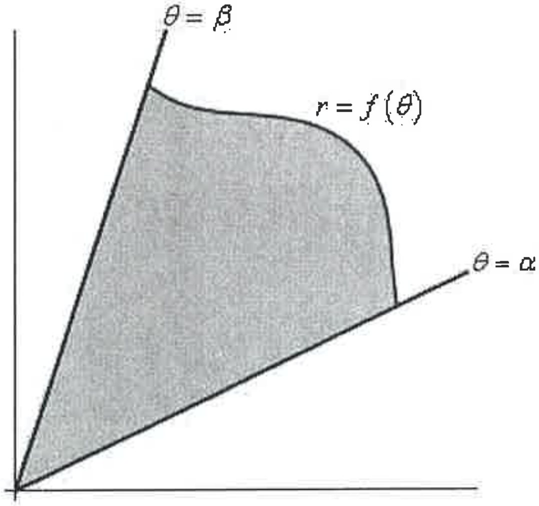
ANSWER KEY

§10.2—Polar Area

We are going to look at areas enclosed by polar curves, that's ***enclosed***, not ***under*** as we typically have in these problems. These problems work a little differently in polar coordinates. Here is a sketch of what the area that we'll be finding in this section looks like.

SECTOR $A=\frac{1}{2}\pi r^2\theta$



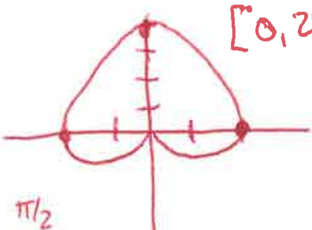


The formula for polar area is different from all previous area formulas, because it is not based on rectangles. Instead, polar area uses and infinite number of sectors to find the area. Remember that a sector is a hunk of a circle, a slice of pizza from the whole pizza.

The area of a sector of a circle is given by $A=\frac{1}{2}\theta r^2$, where θ is in radians. Our area is bounded by the radial lines from $\theta=\alpha$ to $\theta=\beta$

$$Area=\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$$

Example 1:
Find the area bounded by the graph of $r=2+2\sin\theta$.



$[0,2\pi]$

$$2\cdot\frac{1}{2}\int_{-\frac{\pi}{2}}^{\pi/2}(2+2\sin\theta)^2 d\theta$$

$$2\cdot\frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}(2+2\sin\theta)^2 d\theta$$

$$2\cdot\frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}(2+2\sin\theta)^2 d\theta$$

$\sin^2\theta=\frac{1}{2}(1-\cos2\theta)$
$\cos^2\theta=\frac{1}{2}(1+\cos2\theta)$

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}(4+8\sin\theta+4\sin^2\theta)d\theta$$

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}(4+8\sin\theta+2(1-\cos2\theta))d\theta$$

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}(6+8\sin\theta-2\cos2\theta)d\theta$$

$$6\theta-8\cos\theta-2\left(\tfrac{1}{2}\right)\sin2\theta\Big|_{\pi/2}^{3\pi/2}$$

$$\left[6\left(\tfrac{3\pi}{2}\right)-8(0)-\overset{\sin3\pi}{0}\right]-\left[6\left(\tfrac{\pi}{2}\right)-0-0\right]\to 9\pi-3\pi\to\boxed{6\pi}$$

Example 2:
Find the area of one petal of $r = 2\sin 3\theta$.

3 PETALS $[0, \pi]$

FIND POLAR ZEROS

$2\sin 3\theta = 0$

$\sin 3\theta = 0$

$\begin{cases} 3\theta = 0 + 2\pi n \\ 3\theta = \pi + 2\pi n \end{cases}$

$\theta = 0 + \frac{2\pi}{3} n$

$\theta = \frac{\pi}{3} + \frac{2\pi}{3} n$

$\theta = 0, \frac{\pi}{3}$

$$\text{AREA} = \frac{1}{2}\int_0^{\pi/3} (2\sin 3\theta)^2 \, d\theta$$

$$= \frac{1}{2}\cdot 4 \int_0^{\pi/3} \sin^2 3\theta \, d\theta$$

$$= 2\int_0^{\pi/3} \left(\frac{1}{2}\right)(1 - \cos 6\theta)\, d\theta$$

$$= \int_0^{\pi/3} (1 - \cos 6\theta)\, d\theta$$

$$= \theta - \frac{1}{6}\sin 6\theta \Big|_0^{\pi/3}$$

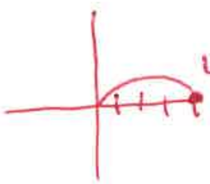
$$\left[\frac{\pi}{3} - \frac{1}{6}(0)\right] - [0 - 0]$$

$$\boxed{\frac{\pi}{3}}$$

ALL 3 PETALS WOULD BE $3\left(\frac{\pi}{3}\right) \to \pi$

Example 3:
Find the area of one petal of $r = 4\cos 2\theta$.

4 PETALS $[0, 2\pi]$



$4\cos 2\theta = 0$

$\cos 2\theta = 0$

$\begin{cases} 2\theta = \frac{\pi}{2} \\ 2\theta = \frac{3\pi}{2} \end{cases}$

$\theta = \pi/4, 3\pi/4$

$$\text{AREA} = 2\left(\frac{1}{2}\right)\int_0^{\pi/4} (4\cos 2\theta)^2 \, d\theta$$

$$= 16\int_0^{\pi/4} \frac{1}{2}(1 + \cos 4\theta)\, d\theta$$

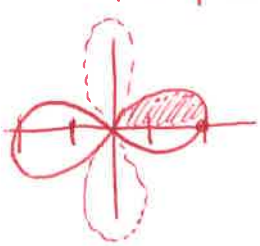
$$= 8\int_0^{\pi/4} (1 + \cos 4\theta)\, d\theta$$

$$= 8\left[\theta + \frac{1}{4}\sin 4\theta\right]\Big|_0^{\pi/4} \to 8\left[\left(\frac{\pi}{4} + 0\right) - (0 + 0)\right] \to \boxed{2\pi}$$

AREA OF ALL 4 PETALS $\to 4(2\pi) \to 8\pi$

Example 4:
Find the area inside one loop of $r^2 = 4\cos 2\theta$.

$r = \sqrt{4\cos 2\theta}$



$4\cos 2\theta = 0$

$\cos 2\theta = 0$

$2\theta = \frac{\pi}{2}$

$2\theta = \frac{3\pi}{2}$

$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$

$$\text{AREA} = 2\cdot\frac{1}{2}\int_0^{\pi/4} (4\cos 2\theta)\, d\theta$$

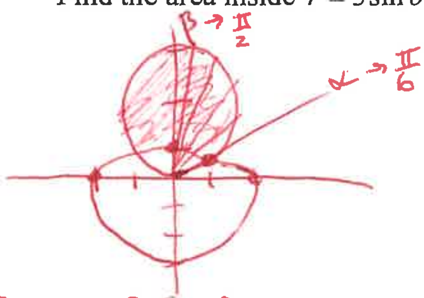
$$= 4\left[\frac{1}{2}\sin 2\theta\right]\Big|_0^{\pi/4}$$

$$= 2[1 - 0]$$

$$= 2$$

Example 5: [0, π] [0, 2π]

Find the area inside $r = 3\sin\theta$ and outside $r = 2 - \sin\theta$.



$$\text{AREA} = 2\left(\tfrac{1}{2}\right)\int_{\pi/6}^{\pi/2}\left[(3\sin\theta)^2 - (2-\sin\theta)^2\right]d\theta$$

USE CALC

$$= \boxed{5.196}$$

$3\sin\theta = 2 - \sin\theta$

$4\sin\theta = 2$

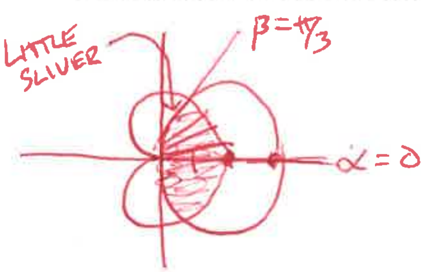
$\sin\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6} \rightarrow \alpha$

$\beta = \frac{\pi}{2}$

Example 6: [0, π] [0, 2π]

Find the area of the common interior of $r = 3\cos\theta$ and $r = 1 + \cos\theta$.



$$\text{AREA} = \cancel{2\left(\tfrac{1}{2}\right)}$$

$$= 2\left[\frac{1}{2}\int_{0}^{\pi/3}(1+\cos\theta)^2\,d\theta + \frac{1}{2}\int_{\pi/3}^{\pi/2}(3\cos\theta)^2\,d\theta\right]$$

$$= \boxed{3.927}$$

$3\cos\theta = 1 + \cos\theta$

$2\cos\theta = 1$

$\cos\theta = \frac{1}{2}$

$\theta = \frac{\pi}{3}$

$r = \theta$

Example 7: CALCULATOR

A polar curve is defined by the equation $r = \theta + \sin 2\theta$ for $0 \le \theta \le \pi$.

(a) Find the area bounded by the curve and the x-axis.

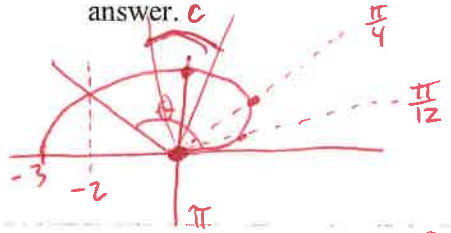
(b) Find the angle θ that corresponds to the point on the curve where $x = -2$.

(c) For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$, $\frac{dr}{d\theta}$ is negative. What does this say about the graph on this interval?

(d) At what angle θ in the interval $0 \le \theta \le \frac{\pi}{2}$ is the curve farthest away from the origin. Justify your answer.

MAXIMIZE r ON $\theta \in [0, \frac{\pi}{2}]$

a)



$$\text{AREA} = \frac{1}{2}\int_0^{\pi} (\theta + \sin 2\theta)^2 \, d\theta$$

$= 4.382$

b) $x = r\cos\theta$

$\frac{\pi}{2}$

π

$x = (\theta + \sin 2\theta)\cos\theta = -2$

Y1 Y2

$\theta = 2.786$

c) IF $\frac{dr}{d\theta}$ IS NEGATIVE ON $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$, THEN THE GRAPH IS MOVING TOWARDS THE POLE ON THIS INTERVAL

d) $r'(\theta) = \frac{dr}{d\theta} = 1 + 2\cos 2\theta = 0$

$\cos 2\theta = -\frac{1}{2}$

$\begin{cases} 2\theta = \frac{2\pi}{3} \\ 2\theta = \frac{4\pi}{3} \end{cases}$

$\theta = \boxed{\frac{\pi}{3}}, \cancel{\frac{2\pi}{3}}$

$r(0) = 0$

$r(\frac{\pi}{3}) = \frac{\pi}{3} + \frac{\sqrt{3}}{2} \to 1.913$

$r(\frac{\pi}{2}) = \frac{\pi}{2} \to 1.571$

SO, THE GRAPH IS FARTHEST FROM THE ORIGIN AT $\theta = \frac{\pi}{3}$

Now try Question 2 from AP BC 2011B exam.

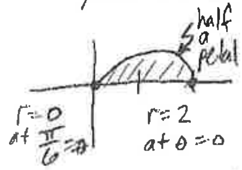
① one petal of $r=2\cos(3\theta)$

polar zeros: $2\cos 3\theta = 0$

$\cos 3\theta = 0$

$3\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$

$\theta = \frac{\pi}{6}, \frac{\pi}{2}$



using symmetry:

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/2}(2\cos(3\theta))^2\,d\theta\right]$$

$$= 4\int_0^{\pi/2}\cos^2(3\theta)\,d\theta$$

*power-reducing ID

$$= 4\left(\tfrac{1}{2}\right)\int_0^{\pi/2}(1+\cos 6\theta)\,d\theta$$

$$= 2\left[\theta + \tfrac{1}{6}\sin 6\theta\right]\Big|_0^{\pi/2}$$

$$= 2\left[\left(\tfrac{\pi}{2}+0\right)-(0+0)\right]$$

$$= \boxed{\pi}$$

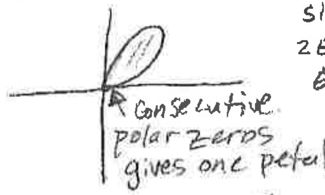
② one petal of $r=4\sin(2\theta)$

polar zeros: $4\sin 2\theta = 0$

$\sin 2\theta = 0$

$2\theta = 0, \pi, \ldots$

$\theta = 0, \frac{\pi}{2}$



$$\text{Area} = \frac{1}{2}\int_0^{\pi/2}(4\sin(2\theta))^2\,d\theta$$

*square & pwr-reducing ID.

$$= 8\left(\tfrac{1}{2}\right)\int_0^{\frac{\pi}{2}}(1-\cos 4\theta)\,d\theta$$

$$= 4\left[\theta - \tfrac{1}{4}\sin(4\theta)\right]\Big|_0^{\pi/2}$$

$$= 4\left[\left(\tfrac{\pi}{2}-0\right)-(0-0)\right]$$

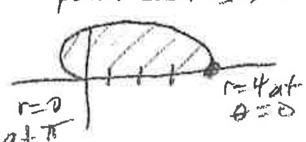
$$= \boxed{2\pi}$$

③ interior of $r = 2+2\cos\theta$

polar zeros: $2+2\cos\theta = 0$

$\cos\theta = -1$

$\theta = \pi$



using symmetry:

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi}(2+2\cos\theta)^2\,d\theta\right]$$

$$= \int_0^{\pi}(4+8\cos\theta+4\cos^2\theta)\,d\theta$$

$$= \int_0^{\pi}(4+8\cos\theta+2(1+\cos 2\theta))\,d\theta$$

$$= \int_0^{\pi}(6+8\cos\theta+2\cos 2\theta)\,d\theta$$

$$= 6\theta + 8\sin\theta + \sin 2\theta\Big|_0^{\pi}$$

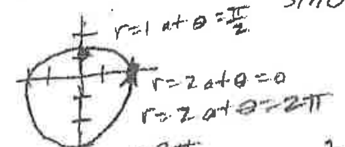
$$= \left[(6\pi+0+0)-(0+0+0)\right]$$

$$= \boxed{6\pi}$$

④ interior of $r = 2-\sin\theta$

polar zeros: $2-\sin\theta = 0$

$\sin\theta = 2 \rightarrow$ no zeros



$$\text{Area} = \frac{1}{2}\int_0^{2\pi}(2-\sin\theta)^2\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}(4-4\sin\theta+\sin^2\theta)\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\left(4-4\sin\theta+\tfrac{1}{2}(1-\cos 2\theta)\right)d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\left(\tfrac{9}{2}-4\sin\theta-\tfrac{1}{2}\cos 2\theta\right)d\theta$$

$$= \frac{1}{2}\left[\tfrac{9}{2}\theta + 4\cos\theta - \tfrac{1}{4}\sin 2\theta\right]\Big|_0^{2\pi}$$

$$= \frac{1}{2}\left[(9\pi+4-0)-(0+4-0)\right]$$

$$= \boxed{\frac{9\pi}{2}}$$

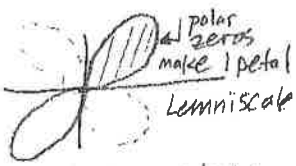
(5) interior of $r^2 = 4\sin(2\theta)$

polar zeros: $4\sin 2\theta = 0$

$\sin 2\theta = 0$

$2\theta = 0, \pi$

$\theta = 0, \frac{\pi}{2}$



using symmetry:

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/2} \overbrace{(4\sin(2\theta))}^{r^2 \text{ already}}\,d\theta\right]$$

$$= 4\int_0^{\pi/2} \sin 2\theta\, d\theta$$

$$= -2\cos 2\theta\Big|_0^{\pi/2}$$

$$= -2[\cos\pi - \cos 0]$$

$$= -2[-1-1]$$

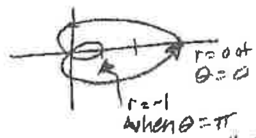
$$= \boxed{4}$$

(6) inner loop $r = 1 + 2\cos\theta$

polar zeros: $1 + 2\cos\theta = 0$

$\cos\theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$ (define inner loop)



$$\text{Area} = \frac{1}{2}\int_{2\pi/3}^{4\pi/3} (1 + 2\cos\theta)^2\, d\theta$$

$$= \frac{1}{2}\int_{2\pi/3}^{4\pi/3} (1 + 4\cos\theta + 4\cos^2\theta)\, d\theta = \frac{1}{2}\int_{2\pi/3}^{4\pi/3} \left(1 + 4\cos\theta + 4\left(\tfrac{1}{2}\right)(1 + \cos 2\theta)\right) d\theta$$

$$= \frac{1}{2}\int_{2\pi/3}^{4\pi/3} (3 + 4\cos\theta + 2\cos 2\theta)\, d\theta$$

$$= \frac{1}{2}\left[3\theta + 4\sin\theta + \sin 2\theta\right]\Big|_{2\pi/3}^{4\pi/3}$$

$$= \frac{1}{2}\left[\left(4\pi + 4\sin\frac{4\pi}{3} + \sin\frac{8\pi}{3}\right) - \left(2\pi + 4\sin\frac{2\pi}{3} + \sin\frac{4\pi}{3}\right)\right]$$

$$= \frac{1}{2}\left[\cancel{4\pi} - 2\sqrt{3} + \cancel{\frac{\sqrt{3}}{2}} - 2\cancel{\pi} - 2\sqrt{3} + \cancel{\frac{\sqrt{3}}{2}}\right]$$

$$= \frac{1}{2}\left[2\pi - 4\sqrt{3} + \sqrt{3}\right] = \frac{1}{2}\left[2\pi - 3\sqrt{3}\right]$$

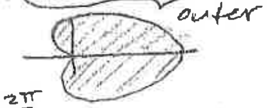
$$= \boxed{\pi - \frac{3\sqrt{3}}{2}}$$

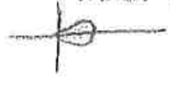
(7) between loops of $r = 1 + 2\cos\theta$

* same curve as in #6.

using symmetry:

$$\text{Area} = \underbrace{2\left[\frac{1}{2}\int_0^{2\pi/3} (1 + 2\cos\theta)^2\, d\theta\right]}_{\text{outer}} - \underbrace{\left(\pi - \frac{3\sqrt{3}}{2}\right)}_{\text{inner (from \#6)}}$$





$$= \int_0^{2\pi/3} (1 + 4\cos\theta + 4\cos^2\theta)\, d\theta - \pi + \frac{3\sqrt{3}}{2}$$

$$= \int_0^{2\pi/3} (3 + 4\cos\theta + 2\cos 2\theta)\, d\theta - \pi + \frac{3\sqrt{3}}{2}$$

$$= \left[3\theta + 4\sin\theta + \sin 2\theta\right]\Big|_0^{2\pi/3} - \pi + \frac{3\sqrt{3}}{2}$$

$$= \left[\left(2\pi + 4\sin\frac{2\pi}{3} + \sin\frac{4\pi}{3}\right) - (0 + 0 + 0)\right] - \pi + \frac{3\sqrt{3}}{2}$$

$$= 2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{2} - \pi + \frac{3\sqrt{3}}{2}$$

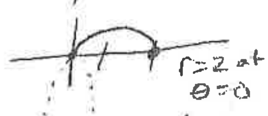
$$= \boxed{\pi + 3\sqrt{3}}$$

(8) one loop of $r^2 = 4\cos(2\theta)$

polar zeros: $\cos 2\theta = 0$

$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$



using symmetry

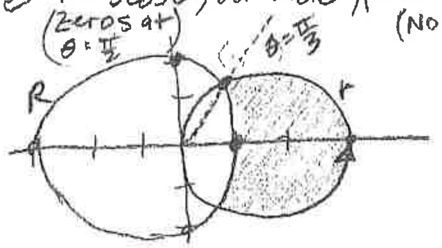
$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/4} \overbrace{(4\cos(2\theta))}^{\text{already squared}}\, d\theta\right]$$

$$= 2\sin(2\theta)\Big|_0^{\pi/4}$$

$$= 2\sin\frac{\pi}{2} - 2\sin 0$$

$$= \boxed{2}$$

(9) inside $r = 3\cos\theta$, outside $R = 2 - \cos\theta$



point of intersection:

$3\cos\theta = 2 - \cos\theta$

$4\cos\theta = 2$

$\cos\theta = \frac{1}{2}$

$\boxed{\theta = \frac{\pi}{3}}$

using symmetry:

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\frac{\pi}{3}} (3\cos\theta)^2\,d\theta - \frac{1}{2}\int_0^{\pi/3} (2-\cos\theta)^2\,d\theta\right]$$

$$= \int_0^{\pi/3} \left[9\cos^2\theta - (4 - 4\cos\theta + \cos^2\theta)\right] d\theta$$

$$= \int_0^{\pi/3} \left[8\cos^2\theta + 4\cos\theta - 4\right] d\theta$$

$$= \int_0^{\pi/3} \left[4(1+\cos 2\theta) + 4\cos\theta - 4\right] d\theta$$

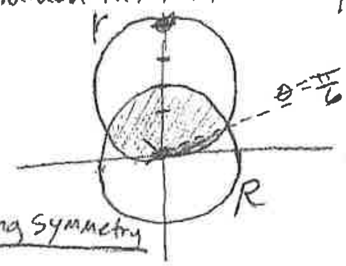
$$= \int_0^{\frac{\pi}{3}} \left[4\cos(2\theta) + 4\cos\theta\right] d\theta$$

$$= 2\sin(2\theta) + 4\sin\theta \Big|_0^{\pi/3}$$

$$= \left(2\sin\left(\frac{2\pi}{3}\right) + 4\sin\left(\frac{\pi}{3}\right)\right) - (0)$$

$$= \sqrt{3} + 2\sqrt{3} = \boxed{3\sqrt{3}}$$

(10) Common interior of $r = 4\sin\theta$, $R = 2$



point of intersect

$4\sin\theta = 2$

$\sin\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6}$

using symmetry

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/6} (4\sin\theta)^2\,d\theta + \frac{1}{2}\int_{\pi/6}^{\pi/2} (2)^2\,d\theta\right]$$

$$= \int_0^{\pi/6} 16\sin^2\theta\,d\theta + \int_{\pi/6}^{\pi/2} 4\,d\theta$$

$$= 8\int_0^{\pi/6} (1-\cos 2\theta)\,d\theta + \int_{\pi/6}^{\pi/2} 4\,d\theta$$

$$= 8\left[\theta - \frac{1}{2}\sin 2\theta\right]\Big|_0^{\pi/6} + 4\theta\Big|_{\pi/6}^{\pi/2}$$

$$= 8\left[\left(\frac{\pi}{6} - \frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)\right) - (0-0)\right] + 4\left[\frac{\pi}{2} - \frac{\pi}{6}\right]$$

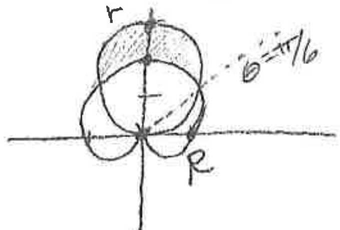
$$= 8\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] + 4\left[\frac{\pi}{3}\right]$$

$$= \frac{4\pi}{3} - 2\sqrt{3} + \frac{4\pi}{3}$$

$$= \boxed{\frac{8\pi}{3} - 2\sqrt{3}}$$

(11) inside $r = 3\sin\theta$, outside $R = 1+\sin\theta$

zero at $\theta = \frac{3\pi}{2}$



point of intersect

$3\sin\theta = 1+\sin\theta$

$2\sin\theta = 1$

$\sin\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6}$

using symmetry:

$$\text{Area} = 2\left[\frac{1}{2}\int_{\pi/6}^{\pi/2}(3\sin\theta)^2\,d\theta - \frac{1}{2}\int_{\pi/6}^{\pi/2}(1+\sin\theta)^2\,d\theta\right]$$

$$= 9\int_{\pi/6}^{\pi/2}\sin^2\theta\,d\theta - \int_{\pi/6}^{\pi/2}(1+2\sin\theta+\sin^2\theta)\,d\theta$$

$$= \int_{\pi/6}^{\pi/2}\left[\frac{9}{2}(1-\cos2\theta) - 1 - 2\sin\theta - \frac{1}{2}(1-\cos2\theta)\right]d\theta$$

$$= \int_{\pi/6}^{\pi/2}\left[3 - 4\cos2\theta - 2\sin\theta\right]d\theta$$

$$= 3\theta - 2\sin2\theta + 2\cos\theta\Big|_{\pi/6}^{\pi/2}$$

$$= \left[\frac{3\pi}{2} - 0 + 0\right] - \left[\frac{\pi}{2} - \sqrt{3} + \sqrt{3}\right]$$

$$= \frac{3\pi}{2} - \frac{\pi}{2}$$

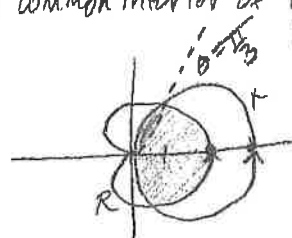
$$= \boxed{\pi}$$

(12) common interior of $r = 3\cos\theta$, $R = 1+\cos\theta$

Polar zero at $x = \frac{\pi}{2}$

zero: $\cos\theta = -1$

$\theta = \pi$



intersection:

$3\cos\theta = 1+\cos\theta$

$2\cos\theta = 1$

$\cos\theta = \frac{1}{2}$

$\theta = \frac{\pi}{3}$

(using symmetry)

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/3}(1+\cos\theta)^2\,d\theta + \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(3\cos\theta)^2\,d\theta\right]$$

$$= \int_0^{\pi/3}\left(1+2\cos\theta+\frac{1}{2}(1+\cos2\theta)\right)d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{9}{2}(1+\cos2\theta)\,d\theta$$

$$= \int_0^{\frac{\pi}{3}}\left(\frac{3}{2}+2\cos\theta+\frac{1}{2}\cos2\theta\right)d\theta + \frac{9}{2}\int_{\pi/3}^{\pi/2}(1+\cos2\theta)\,d\theta$$

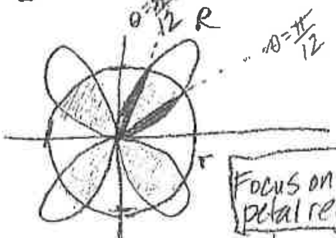
$$= \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin2\theta\right]\Bigg|_0^{\frac{\pi}{3}} + \frac{9}{2}\left[\theta + \frac{1}{2}\sin2\theta\right]\Bigg|_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \left[\left(\frac{\pi}{2} + \sqrt{3} + \frac{\sqrt{3}}{8}\right) - (0)\right] + \frac{9}{2}\left[\left(\frac{\pi}{2}+0\right) - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{4}\right)\right]$$

$$= \frac{\pi}{2} + \sqrt{3} + \frac{\sqrt{3}}{8} + \frac{9\pi}{4} - \frac{3\pi}{2} - \frac{9\sqrt{3}}{8}$$

$$= \boxed{\frac{5\pi}{4}}$$

(13) common interior of $r=4\sin(2\theta)$ & $R=2$



Focus on 1 petal region

zeros:
$\sin 2\theta = 0$
$2\theta = 0, \pi$
$\theta = 0, \frac{\pi}{2}$
(one petal)

intersection
$4\sin 2\theta = 2$
$\sin 2\theta = \frac{1}{2}$
$2\theta = \frac{\pi}{6}, \frac{5\pi}{6}$
$\theta = \frac{\pi}{12}, \frac{5\pi}{12}$

1: Area Contained by 1 Rose Curve:
(using symmetry)

$$\text{Area}_1 = 2\left[\frac{1}{2}\int_0^{\pi/12}(4\sin 2\theta)^2\,d\theta\right]$$
$$= 16\int_0^{\pi/12}\sin^2 2\theta\,d\theta$$
$$= 8\int_0^{\pi/12}(1-\cos 4\theta)\,d\theta$$
$$= 8\left[\theta - \tfrac{1}{4}\sin 4\theta\right]_0^{\pi/12}$$
$$= 8\left[\left(\frac{\pi}{12}-\frac{\sqrt{3}}{8}\right)-(0)\right]$$
$$= \boxed{\frac{2\pi}{3}-\sqrt{3}}$$

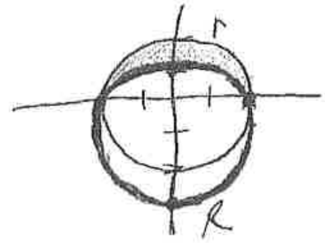
2: Area of 1 sector contained by circle
(using symmetry)

$$\text{Area}_2 = \int_{\pi/12}^{5\pi/12}(2)^2\,d\theta$$
$$= 4\theta\Big|_{\pi/12}^{5\pi/12}$$
$$= 4\left[\frac{5\pi}{12}-\frac{\pi}{12}\right] = \boxed{\frac{4\pi}{3}}$$

Total common Area:

$$\text{Area} = 4\left[\left(\frac{2\pi}{3}-\sqrt{3}\right)+\left(\frac{4\pi}{3}\right)\right]$$
4 regions
$$= 4\left[2\pi - \sqrt{3}\right]$$
$$= \boxed{8\pi - 4\sqrt{3}}$$

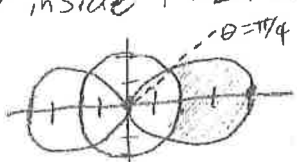
(14) inside $r=2$ & outside $R=2-\sin\theta$



(using symmetry)

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/2}(2)^2\,d\theta - \frac{1}{2}\int_0^{\pi/2}(2-\sin\theta)^2\,d\theta\right]$$
$$= \int_0^{\pi/2}\left[4-(4-4\sin\theta+\sin^2\theta)\right]d\theta$$
$$= \int_0^{\pi/2}\left[4\sin\theta - \tfrac{1}{2}(1-\cos 2\theta)\right]d\theta$$
$$= \int_0^{\pi/2}\left[4\sin\theta - \tfrac{1}{2} + \tfrac{1}{2}\cos 2\theta\right]d\theta$$
$$= -4\cos\theta - \tfrac{1}{2}\theta + \tfrac{1}{4}\sin 2\theta\Big|_0^{\pi/2}$$
$$= \left(0-\frac{\pi}{4}+0\right)-(-4)$$
$$= \boxed{4-\frac{\pi}{4}}$$

(15) inside $r=2+2\cos(2\theta)$ & outside $R=2$



zeros of r:
$\cos 2\theta = -1$
$2\theta = \pi$
$\theta = \frac{\pi}{2}$

intersect:
$2+2\cos 2\theta = 2$
$\cos 2\theta = 0$
$\theta = \frac{\pi}{4}$

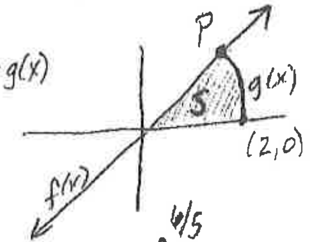
(using symmetry)

$$\text{Area} = 2\left[\frac{1}{2}\int_0^{\pi/4}(2+2\cos(2\theta))^2\,d\theta - \frac{1}{2}\int_0^{\pi/4}(2)^2\,d\theta\right]$$
$$= \int_0^{\pi/4}\left((2+2\cos(2\theta))^2-4\right)d\theta$$
$$= \boxed{5.5707} \text{ (from calculator)}$$

WS 10.2 – Polar Area KEY CALCULUS MAXIMUS

(Dang, this is a LONG worksheet, but totally worth it!)

(16) $y = \frac{2}{3}x = f(x)$, $y = \sqrt{1 - \frac{x^2}{4}} = g(x)$



* x-intercept of g(x)

$1 = \frac{x^2}{4}$, $x^2 = 4$, $\boxed{x=2}$

$\sqrt{1-\frac{x^2}{4}} = \sqrt{\frac{4-x^2}{4}} = \frac{1}{2}\sqrt{4-x^2}$

(a) point of intersection

$\frac{2}{3}x = \sqrt{1-\frac{x^2}{4}}$

$\frac{4}{9}x^2 = 1 - \frac{1}{4}x^2$

$\left(\frac{4}{9}+\frac{1}{4}\right)x^2 = 1$

$x^2 = \frac{36}{25}$

$x = \cancel{-\frac{6}{5}}$ or $\boxed{\frac{6}{5}}$

$y = \frac{2}{3}\left(\frac{6}{5}\right) = \boxed{\frac{4}{5}}$

$\boxed{\text{So Point P is } \left(\frac{6}{5}, \frac{4}{5}\right)}$

(b) $\text{Area} = \int_0^{6/5}\left(\frac{2}{3}x - 0\right)dx + \int_{6/5}^{2}\left(\sqrt{1-\frac{x^2}{4}} - 0\right)dx$

$= \int_0^{6/5}\frac{2}{3}x\,dx + \int_{6/5}^{2}\frac{1}{2}\sqrt{4-x^2}\,dx$ ← need calculator to integrate

$= \boxed{0.9272956436}$

(c) Polar eq. for curve C

$y = \sqrt{1-\frac{1}{4}x^2}$, $y = r\sin\theta$, $x = r\cos\theta$, $x^2+y^2=r^2$

$y^2 = 1 - \frac{1}{4}x^2$

$x^2 + 4y^2 = 4$

$(r\cos\theta)^2 + 4(r\sin\theta)^2 = 4$

$r^2\cos^2\theta + 4r^2\sin^2\theta = 4$

$\boxed{r^2 = \frac{4}{\cos^2\theta + 4\sin^2\theta}}$

or $\boxed{r = \frac{2}{\sqrt{\cos^2\theta + 4\sin^2\theta}}}$

(d) $y = \frac{2}{3}x$

$r\sin\theta = \frac{2}{3}(r\cos\theta)$

$\tan\theta = \frac{2}{3}$

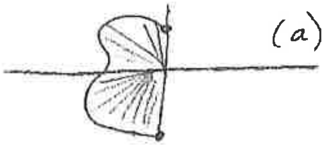
$\boxed{\theta = 0.588 \text{ rads} = A}$

$\text{Area} = \frac{1}{2}\int_0^{A}\left(\frac{4}{\cos^2\theta + 4\sin^2\theta}\right)d\theta$

$= \boxed{0.9272952180}$

* note answers from (b) & (d) are equivalent to 5 decimals only because of the different ways the calculator numerically computes the integrals.

⑦ $r = \theta + \cos(3\theta)$, $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$



(a) $\text{Area} = \frac{1}{2}\int_{\pi/2}^{3\pi/2} (\theta + \cos 3\theta)^2 \, d\theta = \boxed{1.610}$

(b) $y = -1$

$y = r\sin\theta = (\theta + \cos 3\theta)\sin\theta = -1$

* from calculator

$\boxed{\theta = 3.484 \text{ or } \theta = 3.485}$

(c) $\frac{dr}{d\theta} = 1 - 3\sin(3\theta) = 0$

* from calculator

$\theta = 2.207 = A,\ 3.028 = B,\ 4.302 = C$

$\frac{dr}{d\theta} > 0$ for $\theta \in [\frac{\pi}{2}, 2.207) \cup (3.028, 4.302)$. On these intervals the radius is increasing with respect to θ, which means the curve is moving away from the pole/origin.

(d) the distance from the origin $= r$, so we want to maximize r on $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$

$\frac{dr}{d\theta} = 1 - 3\sin(3\theta) = 0$

at $\theta = A, B, C$ (from part (c)). these are the critical values.

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ are the endpts.

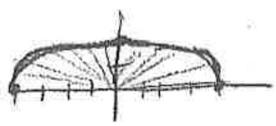
* using the EVT and the closed interval argument.

from calculator

$$\underbrace{\begin{cases} r(A) = 3.150 \\ r(B) = 2.085 \\ r(C) = 5.244 \\ r(\frac{\pi}{2}) = 1.570 \\ r(\frac{3\pi}{2}) = 4.712 \end{cases}}_{\text{Justification}}$$

So the curve is the greatest distance from the origin when $\theta = 4.302$ radians. At this time it is 5.244 units from the origin

(18) $r = \dfrac{4}{1+\sin\theta}$, $0 \le \theta \le \pi$



(a) $\text{Area} = \frac{1}{2}\int_0^{\pi}\left(\frac{4}{1+\sin\theta}\right)^2 d\theta$

$= \boxed{10.667 = \frac{32}{3}}$

(b) $r = \dfrac{4}{1+\sin\theta}$

$r = \dfrac{4}{1+\frac{y}{r}}\left(\frac{r}{r}\right)$

$\dfrac{\cancel{r}}{1} = \dfrac{4\cancel{r}}{r+y}$

$r + y = 4$, $\sqrt{x^2+y^2} = 4 - y$

$x^2 + \cancel{y^2} = 16 - 8y + \cancel{y^2}$

$\boxed{8y = 16 - x^2}$

(c) $y = \frac{1}{8}\left[16 - x^2\right]$

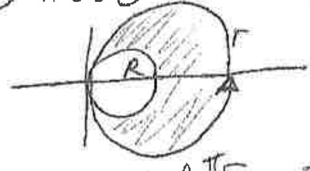
$y = 2 - \frac{1}{8}x^2$, x-ints at $x = -4, 4$

$\boxed{\text{Area} = \int_{-4}^{4}\left(2 - \tfrac{1}{8}x^2\right)dx}$

or using symmetry

$\text{Area} = 2\int_0^4\left(2 - \frac{1}{8}x^2\right)dx$

(19) inside $r = 2\cos\theta$, outside $R = \cos\theta$



1 revolution for $\theta \in [0, \pi]$

$\text{Area} = \frac{1}{2}\int_0^{\pi}\left[2\cos\theta\right]^2 d\theta - \frac{1}{2}\int_0^{\pi}\left[\cos\theta\right]^2 d\theta$

$= \frac{1}{2}\int_0^{\pi}\left[4\cos^2\theta - \cos^2\theta\right]d\theta$

$= \frac{1}{2}\int_0^{\pi} 3\cos^2\theta \, d\theta$

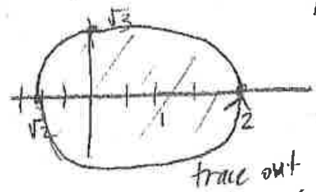
$= \frac{3}{2}\int_0^{\pi}\cos^2\theta \, d\theta$ answer not there

So... using symmetry

$= 2\left[\frac{3}{2}\int_0^{\pi/2}\cos^2\theta \, d\theta\right]$

$= \boxed{3\int_0^{\pi/2}\cos^2\theta \, d\theta}$ $\boxed{A}$

(20) $r = \sqrt{3 + \cos\theta}$



trace out once from 0 to 2π

$\text{Area} = \frac{1}{2}\left[\int_0^{2\pi}\left(\sqrt{3+\cos\theta}\right)^2 d\theta\right]$

$= \frac{1}{2}\int_0^{2\pi}(3 + \cos\theta)\,d\theta$

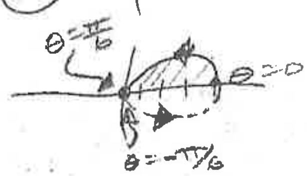
*using symmetry

$= 2\left[\frac{1}{2}\int_0^{\pi}(3+\cos\theta)\,d\theta\right]$

$= \boxed{\int_0^{\pi}(3 + \cos\theta)\,d\theta}$

$\boxed{D}$

(21) one petal of $r = 4\cos(3\theta)$



polar zeros:

$\cos 3\theta = 0$

$3\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

$\theta = \frac{\pi}{6}, \frac{\pi}{2}$

From answer choices:

$A = \frac{1}{2}\left[\int_{-\frac{\pi}{6}}^{\pi/6}\left(4\cos 3\theta\right)^2 d\theta\right]$

$= \boxed{8\int_{-\pi/6}^{\pi/6}\cos^2 3\theta \, d\theta}$ $\boxed{E}$

Finished already! ☺